



Investigations of rock anchor corrosion and its influence factors by exhumations in four typical field sites



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ABSTRACT

With the abundance application of rock and soil anchors in high slope and underground projects, more concerns have been paid to the working condition and durability of the anchorage system. To effectively evaluate the corrosion of anchors in service and the influence factors, field exhumations were performed in four typical application sites and tests of samples were conducted in laboratory. Chemical composition, metallurgical structure and mechanical performance of anchors are tested. Microstructure and component of corrosion byproducts and grout were analyzed. The environmental influence factors are also assessed. The results demonstrate that the anchor head is most liable to corrode in comparing to other section of the anchor. Presence of damaging ions can accelerate the initiation and development of corrosion. Chemical and mechanical properties of anchors remain basically unchanged after 1.5 to 20 years' service. Groundwater and stray currents in the underground do not induce noticeable influence on the corrosion of anchors. This research accomplished a set of evaluation methods of influencing factors and the general law of anchors' corrosion in field environments, which may provide a reference framework for evaluating the environmental effects, durability and service life of rock anchors.

1. Introduction

As an important branch of geotechnical engineering, anchorage technology is widely applied in energy, transportation, hydraulic engineering and city construction fields and, plays a significant role in numerous large projects, because of its effectiveness and economic efficiency. Working state and long-term durability of anchors has been increasingly brought into focus due to the complexity of their working environment and the concealment of underground construction [1,2]. Testing and evaluating the reliability and safety of anchorage structures in a scientific manner and assessing their service life accurately is problems to be settled urgently.

Studies of the safety and durability of anchorage structures are either conducted from a theoretical and experimental perspective or oriented towards field investigations. For the field investigation, with increasing concern about the corrosion damage to rock and soil anchors and of their long-term service life, many efforts have been made by researchers and engineers. The corrosion caused mechanical properties variation of the steel bar or strand is a key criterion to evaluate the corrosion effects and working state of the anchorage system. Studies based on load measurement and strength analysis suggest that the mechanical properties of the anchors reduced drastically due to the corrosion [3–5]. It has been widely accepted that the corrosion is affected by varied factors, including the corrosion resistance design of the anchorage and the environmental effects. Some efforts were made to the study of the

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Table 1
Summary of four field exhumations.

Site	Location	Test position	Anchor type	Service time
Shaping-II	Leshan, Sichuan	Longitudinal cofferdam	Strand	3 yrs
Qishan	Xuzhou, Jiangsu	Tunnels	Bar & Strand	1.5–8 yrs
Jinping-I	Liangshan, Sichuan	Underground powerhouse	Bar & Strand	4 yrs
Manwan	Yunxian, Yunnan	Left side slope	Strand	20 yrs

environmental effects that may cause or accelerate the corrosion. It was concluded that the corrosive mass in the groundwater, the mineralogy and the microbial activity have effective contribution to the anchor corrosion [6,7]. On the other hand, the corrosion resistance design and the construction quality of the anchors also have obvious effects on the corrosion. One case is that the grout with highly alkaline and high content of sulphate ions is the main cause that induce the corrosion [8]. The mechanism and types of corrosion can be characterized based on surveys with statistic methods, which can be used to determine the corrosion potential and present protection methodologies [9–12]. Corrosion accompanied with high stress of the prestressed steel is a distinct feature compared with the general metal corrosion. Stress corrosion cracking is a reason that cause the failure of an anchor in a corrosive environment [13]. The stress corrosion cracking process may deplete the damage tolerance of the prestressed structure. In theoretical and experimental studies, the influence of the environmental and material on stress corrosion cracking can be captured with specialized test system [14,15]. The analysis on morphology of fracture surfaces with optical instruments and the diameter and weight loss is generally conducted in the test simultaneously to reveal the failure mechanism and assess the external influence factors [16,17]. Theoretical and experimental studies can probe into the mechanism and laws of physics and chemistry, through which a better understanding can be reached. However, it is impossible to simulate accurately, in experiments, complex and conceal field environments and the anchorage conditions. Field investigations or exhumation of anchors in service are therefore an essential part of a thorough assessment of the state of an anchorage system for studying the durability and reliability of anchorage structures.

In this study, field exhumation was performed at four typical sites, including longitudinal cofferdam of Shaping-II Hydropower Station, tunnels of Qishan Coal Mine, underground powerhouse of Jinping-I Hydropower Station and left abutment slope of Manwan Hydropower Plant, hereinafter referred to as Shaping-II, Qishan, Jinping-I and Manwan (Table 1). Engineering environment of anchorage structures was studied and analyzed, anchorage conditions of anchors were tested, corrosion conditions, physical and mechanical properties and chemical components of anchors were tested and analyzed. Results from this study can be used as a reference for the assessment of the engineering environment, durability and service life of anchorage structures. Four field exhumations will be described and discussed respectively in the next section.

2. Field exhumations

2.1. Shaping-II

Shaping-II Hydropower Station is located 230 m upstream from the convergence of the Dadu and Guanliao rivers. During the construction, the riverbed was dammed with cofferdams and the river is diverted by an open channel. Reserved rock step on the right bank of the dam served as a guide wall between the powerhouse and dam and then worked as the longitudinal cofferdam. The designed service life of the cofferdam is three years, after which it would be removed. To obtain a better understanding of the working state of prestressed rock anchors in the engineering environment, the exhumation of anchors was made during the dismantling of the rock step. The rock step has a height of 20–30 m, a bottom width of 51 m and a top width of 28 m. The slope ratio is 0.3, as shown in Fig. 1. The prestressed anchors are laid along with the diversion channel with a layout distance of 5.0 m.

The anchors of the Shaping-II are unbonded and prestressed with the steel strand specification being $1 \times 7-15.20-1860$ (GB/T 5224-2014 Steel strands for prestressed concrete) [18]. Each anchor is composed of seven strands and tensioned with a total prestressing force of 1000 kN that lead to an average stress for each strand of 1020 MPa. The borehole diameter is 130 mm. All anchors are totally wrapped by a corrugated pipe with a diameter of 110 mm. Average length of anchors is 35 m. The bond length is 6 m.

According to field conditions, anchors selected for exhumation were cleared and extracted after blasting was conducted in the

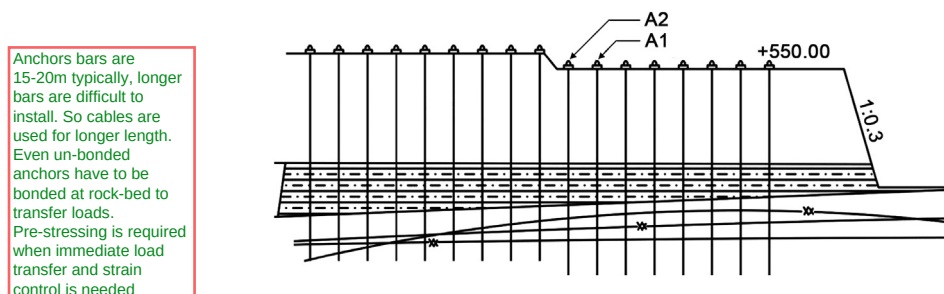


Fig. 1. Prestressed anchors installed in longitudinal cofferdam of Shaping-II Hydropower Station.

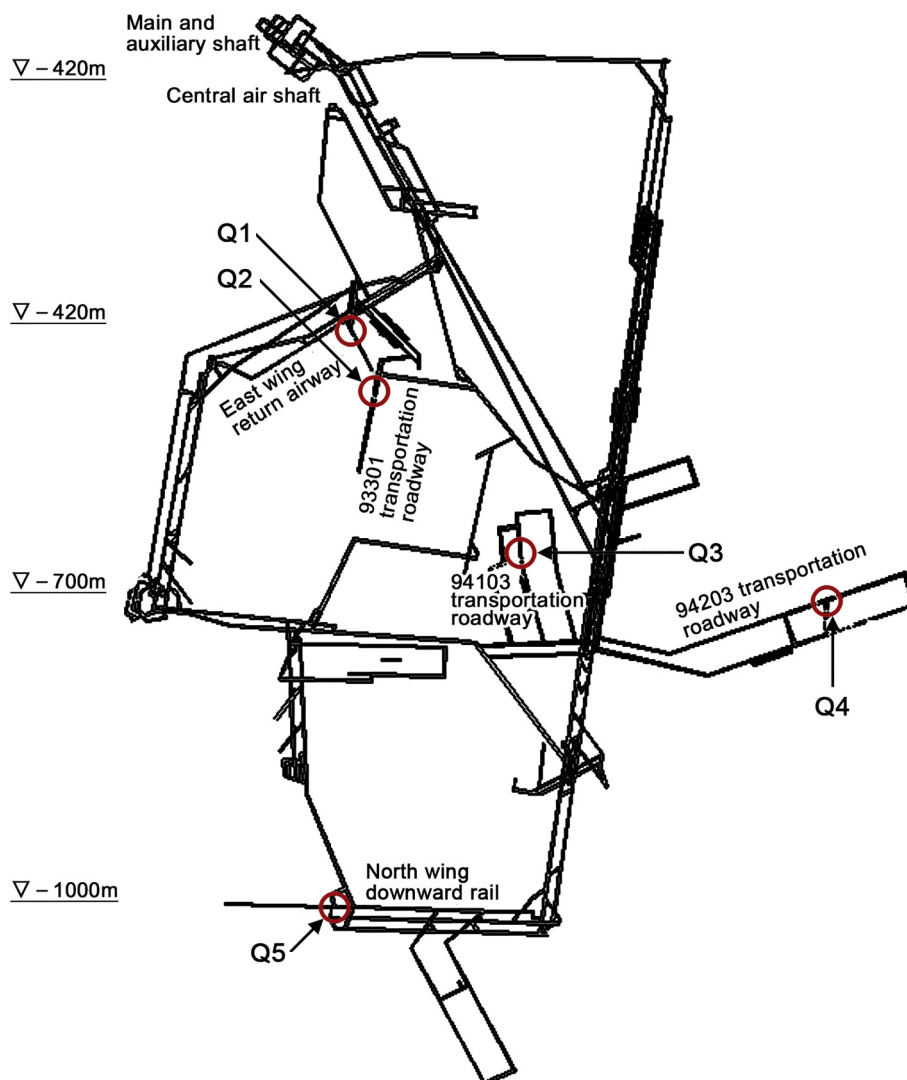


Fig. 2. Sampling locations of Qishan exhumation.

surroundings. Based on field and construction conditions, two neighboring anchors A1 and A2 in the middle of the cofferdam were selected, and two steel strands from each anchor were exhumed and sampled. Both strands were cut into 1 m samples and total 118 strand samples were obtained.

2.2. Qishan

The Qishan site is a coal mine where some anchorage systems are extracted during the exploitation process and others are in service. Anchors were exhumed from five locations and their in-service age varied from 1.5 to 8 years. At this site, **drilling was conducted with a rotary hydraulic jumbolter** which allows adjustment of the drill angle and replacement with a bigger drill bit during

Table 2
Anchor parameters of Qishan exhumation.

Sample	Tunnel name	Depth (m)	Bar diameter(mm)	Length (mm)	Pullout resistance (kN)	Service time
Q1	East wing return airway	700	20	2400	≥80	5.0 yrs
Q2	93,301 transportation roadway	730	16	2000	≥80	2.0 yrs
Q3	94,103 transportation roadway	820	20	1800	≥80	1.5 yrs
Q4	94,203 transportation roadway	850	20	2400	≥80	3.0 yrs
Q5	North wing downward rail	1000	18	2400	≥80	8.0 yrs

the drilling [19].

Sampling locations are illustrated in Fig. 2. Design parameters of the anchors in these locations are summarized in Table 2. During exhumation, seepage was observed at all locations.

2.3. Jinping-I

The large-scale chamber groups of the underground powerhouse of the Jinping-I Hydropower Station are composed of the diversion tunnel, the main powerhouse, the busbar tunnel, the primary transformer chamber, the tailrace surge chamber and the tailrace chamber. The geological conditions are very complex in this area. The surrounding rock is weak and in a high ground stress area with many faults. Under high ground stress, excavation and unloading of chambers lead a redistribution of surrounding rock stress, accompanied with an evolution of rock mass structure and macro-mechanical damage and degradation. This causes unloading efficiency damage and crack propagation, deformation and enlargement of surrounding rock of wide-span and high-side-wall chambers. As a result, the stress of anchors and other supporting structures keeps increasing. Some of them even exceeding the designed bearing capacity.

In the underground powerhouse with complicated geological conditions, installed anchors will face possible corrosive materials existed in the fractured rock mass, wetting and drying cycles, temperature and humidity, stray currents, bimetal function, and other factors. These factors will result in continuous corrosion of anchors.

In the underground powerhouse chambers, surrounding rock is reinforced with anchoring and shotereting technology for permanent support. Anchors used here include grout bolts, self-propelled bolts, prestressed bolts and steel strands. Anchors of the main powerhouse include grout bolts with 28 mm in diameter and 6 m in length, which are installed every 3 m. Prestressed bolts with 32 mm in diameter, 9 m in length and load capacity of 120 kN are installed every 2.4 m. The average stress of the bolts is 149 MPa. Prestressed steel strands adopted are 20 or 25 m long, with a load capacity of 1000 kN, 1750 kN, 2000 kN and 2500 kN for different locations, and are arranged in a staggered manner. The average stress for each strand of these anchors is 1020, 1250, 1099 and 1050 MPa correspondingly.

Field measurements were performed at Jinping-I. Specifically, temperature, humidity, water sample, pH and electrical stray current of underground caverns were evaluated.

2.4. Manwan

Anchor exhumed at the Manwan site is located on the left abutment slope of Manwan Power Plant (Lincang City, Yunnan Province). This anchor is composed of 8 steel strands which are fully bonded with a total prestressed load of 1000 kN and average stress of 893 MPa for each strand. The steel strand is 25 m long with yielding strength of 1470 MPa. The bond length is 6.5 m. The diameter of the borehole is 115 mm. During exhumation, crowded drilling and coring were performed by machines to generate a porous structure with lower strength. Tunneling was then carried out as rocks were unleashed artificially with air picks [20].

The holes are bored around the bolt to make the rock weaker, then using the air picks the rock is broken without damaging the anchor

3. Field investigation results

In the 4 exhumation sites presented above, working environment and state of anchor were characterized and the corrosion morphologic characteristics of the anchorage structures were recorded (Table 3).

3.1. Groundwater

For prestressed anchors in the rock and soil, the most fundamental corrosion type is electrochemical corrosion of metals [21]. The corrosion is influenced by numerous environmental factors, among which damaging ion concentration and pH of the water environment are most prominent. Damaging ions and pH of surface water and groundwater in the engineering region were tested in accordance with the Code for Hydropower Engineering Geological Investigation (GB 50287-2006) [22] and the Code for Investigation of Geotechnical Engineering (GB 50021-2001) [23]. According to groundwater corrosiveness test results listed in Table 4 for Shaping-II, the groundwater is alkaline, and its damaging ion concentration falls in the slight corrosion scope, imposing practically no corrosive action on steel strands.

Water leakage of Qishan Coal Mine tunnels is largely due to roof sandstone crack. Test results of pH on groundwater that drains from the test sites (Table 5) show that the water is alkaline and there is no general acidic corrosion.

Table 3
Field investigations performed.

	Shaping-II	Qishan	~~~~~ Jinping-I	1
Groundwater	✓	✓	✓	✓
Temperature		✓	✓	
Relative Humidity		✓	✓	
Underground stray currents			✓	
Morphology	✓	✓	✓	✓

Bicarbonate can for a protective layer and buffer pH but high level can be corrosive, CO_2 dissolves with water to form acid which lowers pH of water, Mg^{++} is not too dangerous, sulphate ions can cause pitting. Low pH dissolves oxides and causes corrosion and high pH can be protective but is not good for grout.

Table 4
Shaping-II groundwater test results.

Item	Test value	Standard value	Description
HCO_3^-	2.01–5.04 mmol/L	> 1.07 mmol/L	No dissolution corrosion
Corrosive CO_2	0–9.0 mg/L	< 15 mg/L	No carbonic acidic corrosion
Mg^{2+}	3–29.1 mg/L	< 1000 mg/L	No magnesium sulfate corrosion
SO_4^{2-}	2.4–97.0 mg/L	< 250 mg/L	No magnesium sulfate corrosion
pH	7.1–10.9	> 6.5	No general acidic corrosion

Table 5
Qishan groundwater test results.

Test site	pH	Standard value	Description
Site 1	8	> 6.5	No general acidic corrosion
Site 2	9	> 6.5	No general acidic corrosion

Two bottles of water were sampled from the drainage gallery at the second floor of the underground powerhouse at Jinping-I. Sample water testing included pH value, ionic concentration of Cl^- and SO_4^{2-} . The water test results listed in Table 6 indicate that the underground powerhouse is slightly alkaline with a low content of deleterious ions. The water corrosivity falls into the slight corrosion scope.

Table 7 gives the water test results of the Manwan exhumation site. Water of the surrounding environment of the steel strands is slightly alkaline with a low average concentration of damage ions. The water doesn't show obvious corrosivity to the anchors.

3.2. Temperature and humidity

In iron and steel, the corrosion rate for a given oxygen concentration approximately doubles for every 30 °C up to 80 °C when corrosion is controlled by diffusion of oxygen [24]. The corrosion rate increases even faster with hydrogen evolution in the corrosion. The moisture of the anchorage structure, affected directly by the atmospheric humidity, is also an important factor that controls the corrosivity of anchors. Therefore, the temperature and humidity around the chosen samples were recorded in the field exhumation.

Humidity was measured as relative humidity (RH), namely the ratio of partial pressure of water vapor to the equilibrium vapor pressure of water at a given temperature.

According to the data collected during the construction and field tests of the Qishan Coal Mine, temperature and relative humidity of exhumation site were recorded as listed in Table 8.

Temperature and relative humidity statistics of the main powerhouse, the primary transformer chamber and the tailrace surge chamber of Jinping-I Hydropower Station are summarized in Table 9. Temperature of the primary transformer chamber is higher than in other cavities, followed by the PD66 exploration adit on the left slope and lower values in the tailrace surge chamber and the main powerhouse. The tailrace surge chamber ranks top by relative humidity, followed by the main powerhouse, and lower in the primary transformer chamber and PD66 exploration adit. For hydropower stations, temperature is usually inversely proportional to relative humidity.

The corrosion occurs with diffusion of oxygen on surface, higher temperature increases diffusion so can cause more corrosion. If there is not much oxygen hydrogen emission can also occur in presence of acidic env which is a diff mech and even more dangerous

3.3. Underground stray current

Stray current, which exist widely in rock and soil mass media, is another influencing factor of anchor corrosion. Anchor corrosion caused by stray currents is recognized as stray current corrosion. It is mainly based on the electrolysis of electrochemical corrosion. Mainly made of steel bars or strands, anchors are electrically conductive. A corrosion cell is produced as stray currents flow in an anchor and thereby, forms a potential difference, in which the metallic part from which the current flows in is called the cathode zone and the one from which the current flows out is called the anode zone. The more intense the stray currents are, the more severe and faster the anchor corrosion will be.

In the stray current test, CuSO_4 solution electrode was taken as the potential sensor. It was placed as close as possible to the rock

Table 6
Jinping-I groundwater test results.

Item	Test value		Standard value	Description
	Sample 1	Sample 2		
Cl^-	0.53 mg/L	0.49 mg/L	< 100 mg/L	Slight corrosion
SO_4^{2-}	9.54 mg/L	9.57 mg/L	< 250 mg/L	No magnesium sulfate corrosion
pH	7.72	7.77	> 6.5	No general acidic corrosion

Table 7
Manwan water test results.

Item	Test value	Standard value	Description
Cl ⁻	2.64 mg/L	< 100 mg/L	Slight corrosion
SO ₄ ²⁻	180 mg/L	< 250 mg/L	No magnesium sulfate corrosion
pH	7.56	> 6.5	No general acidic corrosion

Table 8
Tunnel temperature and humidity of Qishan coal mine.

Test site	Temperature (°C)	Relative humidity (%)
Q1	25.18	93.6
Q2	24.29	82.9
Q3	31	–
Q4	31	–
Q5	38.47	–

Table 9
Temperature and humidity of Jinping-I.

Test location	Average temperature (°C)	Average relative humidity (%)
Main powerhouse	17.8	77.81
Primary transformer chamber	21.4	74.10
Tailrace surge chamber	17.0	88.34
PD66 exploration adit	19.0	73.11

mass around anchor. The testing terminal positioned at the end of the anchor and a wire immersed in the CuSO₄ solution electrode were connected with the voltmeter (Fig. 3).

Stray current tests for anchors in the underground powerhouse and the primary transformer chamber of Jinping-I hydropower Station were conducted. Test data of stray currents for all test points of the main powerhouse are shown in Fig. 4.

At main powerhouse and primary transformer chamber, the potential difference between anchor bolt and rock mass varies with time within a potential range (Fig. 4). Potential difference between anchor and surrounding rock mass is basically stable and slightly fluctuates between test points and rock. In practice, if the potential is not less negative than -0.850 vs. Cu/CuSO₄, then the corrosion can be negligible [25]. It can thus be deemed that there are no stray current effects surrounding anchors.

3.4. Morphologic characteristics

3.4.1. Shaping-II sample

Analysis of ground anchorage failure shows that corrosion damage is the cause in most cases. For prestressed non-bonded anchors, anchor head is an essential component susceptible to rusting. Since prestressed non-bonded anchors do not bear unnecessary constraints in full length, once the anchor head is damaged by rusting, the entire anchor may instantly be broken or failed and thereby stability and safety of the rock may be compromised.

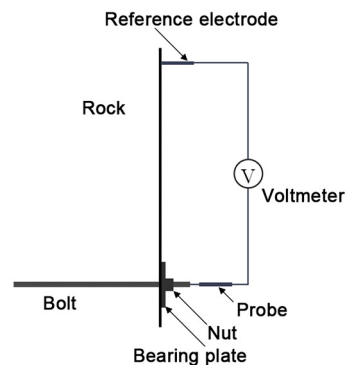


Fig. 3. Stray current test in the field.

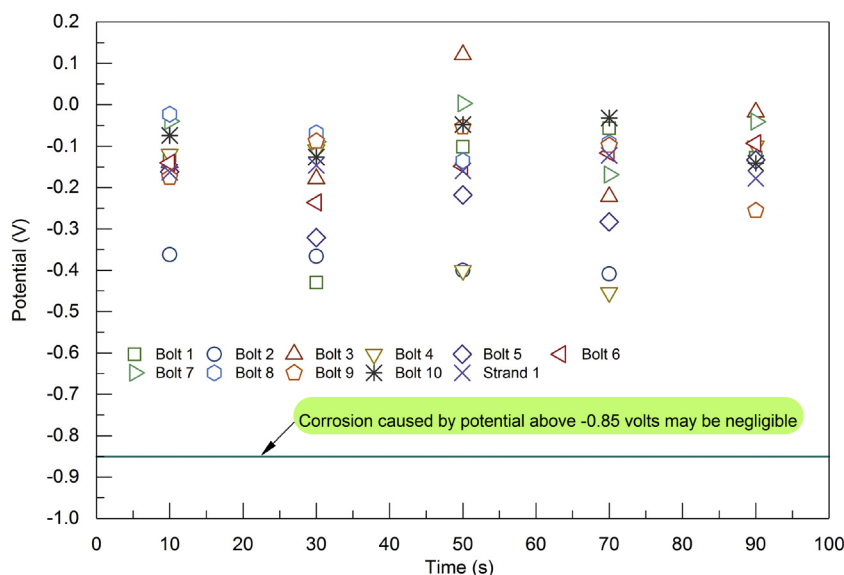


Fig. 4. Stray current testing for anchors in main powerhouse and primary transformer chamber of Jinping-I Hydropower Station.

At the Shaping-II exhumation site, the corrosion of the anchor head and bearing plate can be seen clearly after removing of the concrete wall and part of superficial rock (Fig. 5). The anchor head was covered by abundant yellowish-brown rust. More corrosion was found in the outer side than inner side of the bearing plate. This may be due to the contact of the inner side of the bearing plate and the concrete wall that gives the bearing plate more protection from air and water.

The morphology of the sample from the unbonded length is shown in Fig. 5(b). The steel strand was uniformly coated with grease. The sample displays a smooth surface with brightly black metallic luster after cleaning. No obvious rusting was detected.

The bond length of the anchors in Shaping-II exhumation site is 6.0 m in length. The exhumed sample of the bond length is shown in Fig. 5(c). Rusting was observed after cleaning on the outer side of the steel strand. Its surface was covered with yellowish-brown rust, while the inner side was bright with no noticeable corrosion. Rusting causes may include: (1) low grouting quality during construction may cause steel strand of the bond length to be locally exposed to external air and moisture, (2) water and moisture may reach the anchor through cracks that caused by the explosion and vibration during the exhumation.

According to morphology inspection of the anchor corrosion, compared with Champion standard charts [26], the unbonded length, bond length and the anchor head were determined slightly corroded, corroded, and severely corroded respectively. It may

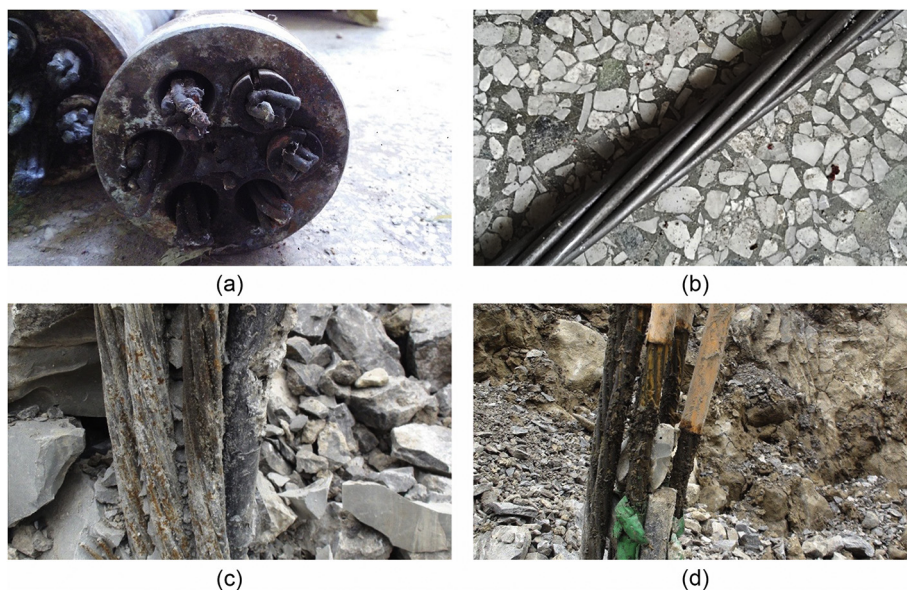


Fig. 5. Morphology of anchor corrosion of Shaping-II exhumation site: (a) anchor head, (b) free section, (c) bonded length, (d) free section to bonded length.

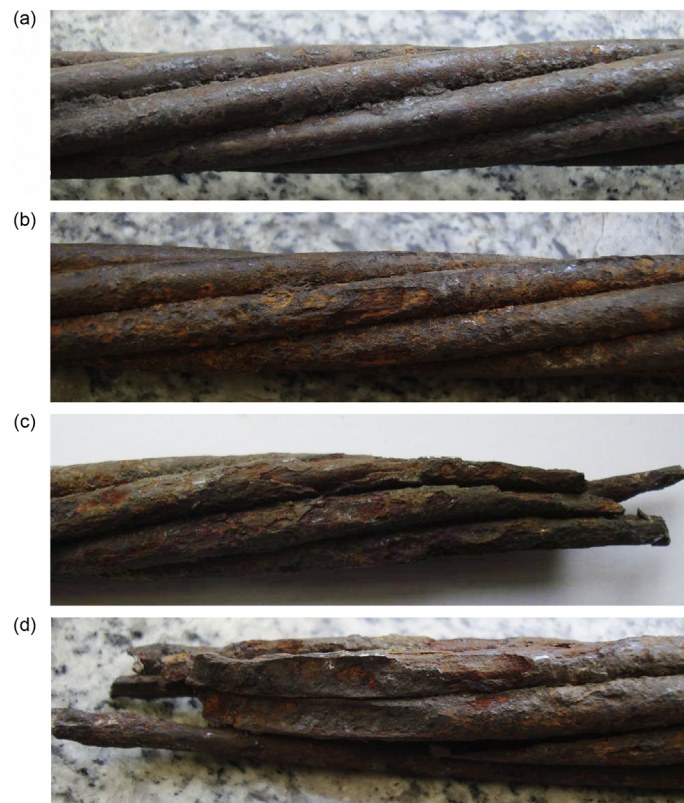


Fig. 6. Morphology of the anchor sample from Qishan: (a) corrosion morphology of steel strand near anchor head, (b) corrosion morphology of steel strand about 90 cm far from anchor head (with obvious pitting corrosion), (c) fracture morphology of the strand on the side connected to the anchor head, (d) fracture morphology of the strand on another side.

conclude that: (1) the anchor head is a critical part liable to corrode with a high need for anti-corrosion measures. (2) the strand sheath filled with grease can provide effective protection for prestressed unbonded anchors. (3) the grouting is an effective means of corrosion resistance for anchors. Therefore, the construction quality process control of grout is very important.

3.4.2. Qishan sample

At the Qishan exhumation site, a macroscopic morphologic analysis was conducted on a failed steel strand sample that had served for 6 years in the coal mine (Fig. 6). Fig. 6(a) is a sample near the anchor head. Fig. 6(b) is a sample about 90 cm far from anchor head. Fig. 6(c) and (d) display the broken section of the strand which is about 120 cm from the anchor head. The corrosion on the surface of the sample near the anchor head shows obvious uniform corrosion and no pitting corrosion was observed. More pitting corrosion was found on the surface of the steel strand 90 cm far from the anchor head, as shown in Fig. 6(b). The most pitting corrosion was found near the fracture, and peeling was even visible at some places (Fig. 6(c) and (d)).

It can be seen from Fig. 6(c) and (d) that each steel wire of the strands was broken at the same location. The break took place along an inclined surface at around 30 degree for most of the wires. Whereas the other wires were broken along the vertical section and presented flat-type fracture. The crack might initiate due to the pitting corrosion and peeling on the surface of the strands and propagate along the inclined surface. The propagation of the crack may reduce the anchors diameter and lead to a stress concentration resulted in a flat-type fracture of the strands when the stress exceeds the tensile strength. Therefore, it can be concluded that the corrosion should be the reason of the initiation of the wire's crack. Then the crack propagated under both the effect of the corrosion and the high stress. The wires broken when the stress beyond the strength. Thus, the fracture should be a result of the combined effects of stress and corrosion.

3.4.3. Jinping-I sample

Field data at the Jinping-I exhumation site include only paper records on corrosion of the anchor head. There are no photographs that would have permitted further morphological analysis. Corrosion characteristics of the anchor head are the most evident corrosion phenomenon that can be observed by the naked eye. The corrosion degree of the anchor bolts could be judged qualitatively based on the observations on corrosion characteristics of the anchor head and a standardized methodology [16]. Therefore, the degree of corrosion of the anchors applied in the main powerhouse of Jinping-I Hydropower Station was determined as moderate corrosion.

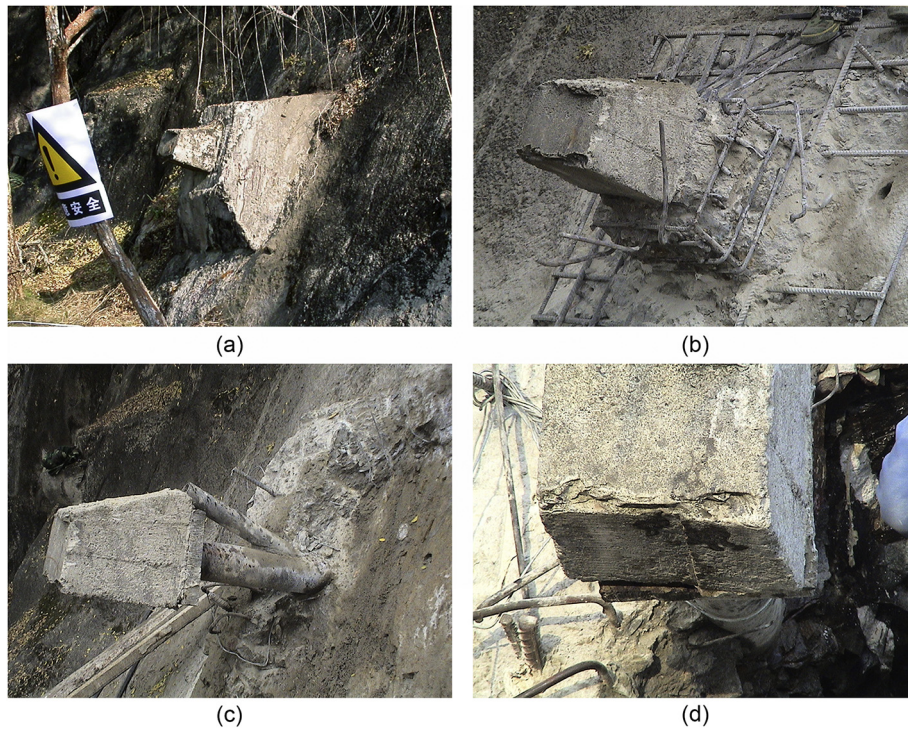


Fig. 7. Anchor pier exhumation procedure at Manwan: (a) exterior appearance, (b) dismantling of anchorage pier, (c) bearing plate and grouting tube, (d) anchor head.

3.4.4. Manwan sample

The anchor head and anchorage pier of the Manwan exhumation site is shown in Fig. 7(a). The reinforced concrete anchorage pier was 1 m in length and width, as shown in Fig. 7(b). Before the exhumation, no cracks were found in the anchorage pier and its surroundings. Dismantling of the anchorage pier unveiled the grout tube and a 4 cm-thick bearing plate, as shown in Fig. 7(c).

Removal of the anchor head covering concrete exposed a rusted anchor head, as shown in Fig. 8(a). The corrosion of the anchor head indicates that a concrete cover cannot prevent corrosion of the anchor head well.

Corrosion of steel strands at different burial depths is different. Steel strand samples are bent, which might be the result of long-term high stress. The concave side of the strand displays morphological corrosion characteristics that are different from those on the convex side (Table 10).

The steel strand surface indicates that the grout thickness varies greatly with anchor burial depth. In the bonded length, grout cover is thicker on the concave side than on the convex side. Steel strands without grout cover are usually covered with brownish-yellow rust or a bluish-grey surface. Basically, there is no grout bonded on the surface of steel strand at a burial depth of 2 m and 11 to 18 m. Some grout is present on the concave side of the steel strand at burial depths of 6 and 7 m. Grout is found on the convex side of steel strand at burial depth of 1 m. In view of the overall situation, corrosion products distributed the whole surface of the steel strand. And in some depth, corrosion products have permeated to the surface of the residual cement mortar. It thus appears that grout



Fig. 8. Corrosion of anchor head of Manwan exhumation: (a) anchor head corrosion, (b) corrosion of wedges and steel strand.

Table 10
Morphology of steel strand surface at Manwan exhumation site.

Burial depth (m)	Surface grout	Corrosion on the surface
1	Convex: filmy grout Concave: no grout	Convex: scattered brownish-yellow rust Concave: bluish-grey rust
2	Virtually no grout	Convex: scattered brownish-yellow rust Concave: bluish-grey rust
6	Convex: no grout Concave: well grouted	Convex: bluish-grey, little brownish-yellow rust Concave: seriously corroded, brownish-yellow rust
7	Convex: no grout Concave: well grouted	Convex: bluish-grey, little brownish-yellow rust Concave: seriously corroded, brownish-yellow rust
11	Virtually no grout	Bluish-grey rust on both sides
12	Virtually no grout	Convex: bluish-grey Concave: some brownish-yellow rust
13	Virtually no grout	Bluish-grey rust on both sides
14	Virtually no grout	Bluish-grey rust on both sides
15	Partially grouted	Convex: bluish-grey, a little brownish-yellow rust Concave: a lot of brownish-yellow rust
16	No grout	Bluish-grey rust on both sides
18	No grout	Convex: bluish-grey, a little brownish-yellow rust Concave: a little brownish-yellow rust

did not provide adequate protection to steel strand. Corrosion is relatively intense at burial depths of 1, 2, 6, 7, 15 and 18 m. The corrosion area on the surface of the steel strand in those depths is larger. Brownish-yellow rust is observed on the center wire of the strand as well as on the concave surface of the strand, while the convex surface of the strand is basically covered with brownish-yellow or bluish-grey rust. In general, the bonded length of the anchor has higher corrosion resistance because of the grout protection. The free length of the anchor with poor grout or without grout shows more serious corrosion.

4. Laboratory test results

Laboratory tests were conducted on steel bar and strand samples from field exhumation at Shaping-II, Qishan and Manwan sites. The tests performed included chemical composition analysis, metallurgical structure analysis, electrochemical analysis, mechanical analysis, corrosion byproducts analysis, microhardness analysis, and grout analysis (Table 11).

4.1. Chemical composition analysis

Chemical properties of metals are mainly characterized by heat and corrosion resistance. In a complex ground environment, corrosion resistance plays an especially prominent role on performance of prestressed anchors. Behavior of metallic corrosion basically rests on the chemical composition of the metal since it significantly affects the metallurgical structure and corrosion resistance.

As shown in Table 12, chemical composition analysis on anchor samples from the Shaping-II exhumation site was found meeting the chemical composition requirements of No. 80 steel in the Quality Carbon Structural Steels (GB/T699-1999) standard [27].

Based on the original records of Manwan Hydropower Plant, the steel strands and their composition conform to the requirements of prestressed steel strand stated in GB5224-85 [28] and the steel composition requirements stated in GB699-88 [29]. Table 13 shows the chemical composition tests results of anchor samples from the Manwan exhumation site. The results indicate that after 20 years' service, the chemical composition of the exhumed steel strands remains unchanged and meets the requirements of No. 80 steel stated in GB/T699-1999.

4.2. Metallurgical structure analysis

Although the metallurgical structure of steel strand is relatively stable after smelting and casting, it also can be changed in the

Table 11
Laboratory tests performed.

Tests	Shaping-II	Qishan	Manwan
Chemical composition	✓		✓
Metallurgical structure	✓	✓	✓
Electrochemical analysis	✓		
Mechanical analysis	✓		✓
Corrosion byproduct analysis		✓	✓
Microhardness analysis		✓	
Grout Component analysis			✓

Table 12
Chemical composition of Shaping-II steel strand samples.

Determined element	Measured value (%)	Standard value (%)
C	0.83	0.77–0.85
Si	0.22	0.17–0.37
Mn	0.80	0.50–0.80
P	0.018	≤ 0.035
S	0.0049	≤ 0.035
Cr	0.20	≤ 0.25
Cu	0.011	≤ 0.25
Ni	0.0076	≤ 0.30

Table 13
Chemical composition of Manwan steel strand samples.

Determined element	Measured value	Standard value
C	0.85	0.77–0.85
Si	0.25	0.17–0.37
Mn	0.68	0.50–0.80
P	0.014	≤ 0.035
S	0.034	≤ 0.035
Cr	0.024	≤ 0.25
Cu	0.058	≤ 0.25
Ni	0.022	≤ 0.30

thermal treatment and mechanical processing. This change will affect the corrosion resistance and mechanical performance of materials.

A metallurgical analysis was performed on the steel strand samples at the unbonded length of the anchors exhumed at Shaping-II site. In view that the high heat generated by cutting during the sampling may influence the metallurgical structure of the steel strand, small segments for microscopic metallurgical morphology analysis were obtained from the mid and end parts of the samples (Fig. 9). Depending on the results, metallurgical structures of the end and mid parts of the steel strand are both cold-deformed sorbites, consistent with the metallurgical structures of factory materials. The heat generated by the cutting didn't affect the metallurgical structure and should thus produce no change in mechanical properties.

The metallurgical analysis was conducted on steel strand samples from Qishan site. The distribution of nonmetallic inclusions and the metallurgical structure of the samples was tested. Morphology and EDS (Energy Dispersive Spectrometer) analysis results of nonmetallic inclusions of the polished longitudinal sample are shown in Fig. 10. Results of nonmetallic inclusions are shown in Table 14. According to the results of the nonmetallic inclusion examination, nonmetallic inclusion of the sample mainly includes thick and fine Type B (aluminate oxide) and Type D (globular oxide) inclusion at severity levels of 0.5 for all cases [30, 31].

Cross-section and longitudinal samples from Qishan exhumation site were shaped and polished into metallurgical samples and etched with a 4% nital solution. Metallurgical structures were then observed under a metallurgical microscope (Figs. 11 and 12). Samples have an even metallurgical structure. Both cross-section and longitudinal samples are formed of tempered sorbite.

Metallurgical structures of cross-section and longitudinal sections of the steel strand samples buried for 20 years from the Manwan exhumation are shown in Figs. 13 and 14.

The results show that the metallurgical structure remains unchanged for sections of different wires in a strand at various burial depths. The environment and high stress do not have obvious effect on the metallurgical structure of steel wires.

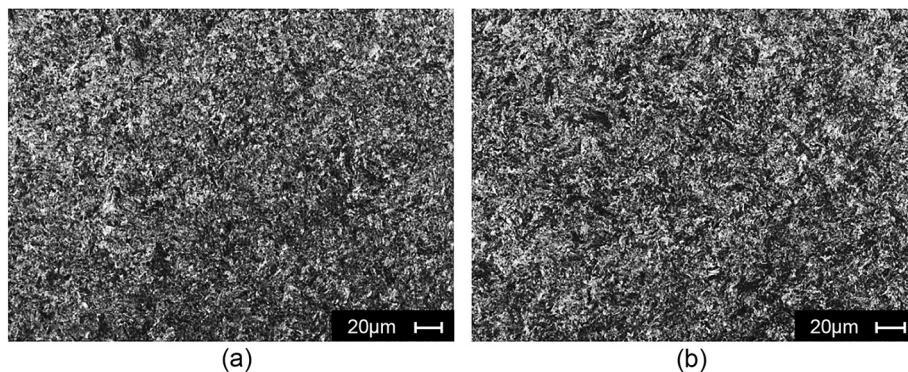


Fig. 9. Metallurgical structure of steel strand samples from Shaping-II exhumation: (a) end of the sample, (b) mid of the sample.

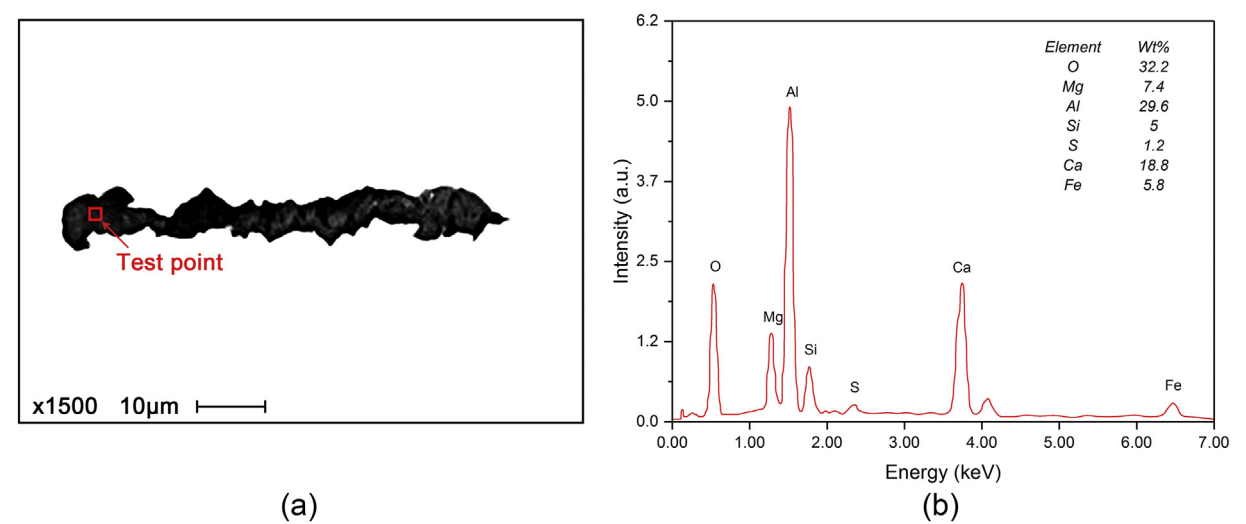


Fig. 10. Morphology and EDS analysis of inclusions of longitudinal steel strand sample from Qishan coal mine: (a) morphology of the inclusions, (b) EDS result.

Table 14
Nonmetallic inclusion and metallurgical test results of Qishan coal mine samples.

Inclusions	Inclusions							
	Type A		Type B		Type C		Type D	
	Thick	Thin	Thick	Thin	Thick	Thin	Thick	Thin
Severity level	–	–	0.5	0.5	–	–	0.5	0.5

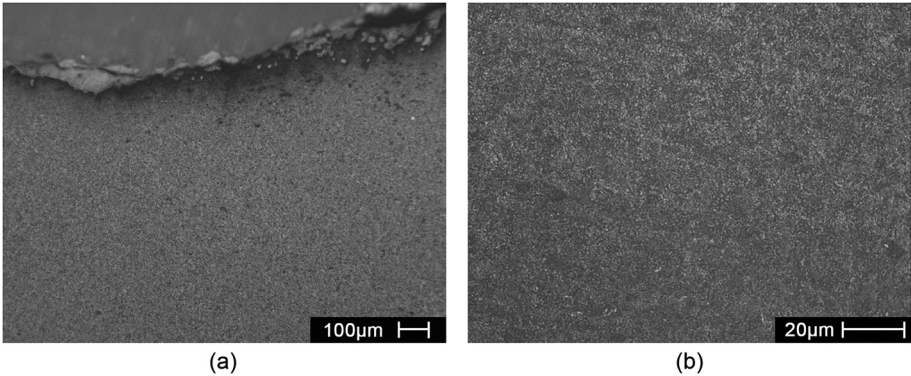


Fig. 11. Metallurgical structure of cross-section sample of steel strand from Qishan exhumation: (a) 100 magnification, (b) 1000 magnification.

4.3. Electrochemical test

Electrochemical test is an approach for the study of metal corrosion by analyzing the relation between electrode potential and polarization current or polarization current density. This method can be used to analyze the corrosion sensitivity and rate, and thereby serve as the basis for evaluating corrosion resistance of anchors. The tests were conducted by an electrochemical workstation (Princeton, 273A) in this research.

A polarization curve test was conducted on the steel strand from Shaping-II exhumation site (Figs. 15, 16 and 17). Table 15 summarizes pH values and Cl⁻ concentrations of solutions used for polarization curve analysis of a steel strand. As shown in Fig. 17, compared to Figs. 15 and 16, the slope of the Tafel line for saturated Ca(OH)₂ solution is higher than that of other two solution. This means to acquire the same current density increasing, more overpotential is needed. This infers that the sample in the saturated Ca (OH)₂ solution has higher corrosion resistance. The corrosion rate can be calculated based on the current density with the equation $V = A \cdot I_{\text{corr}} / (n \cdot F \cdot \rho)$, where A is the atomic weight, I_{corr} is the current density, n is the lost electron number, F is the Faraday constant

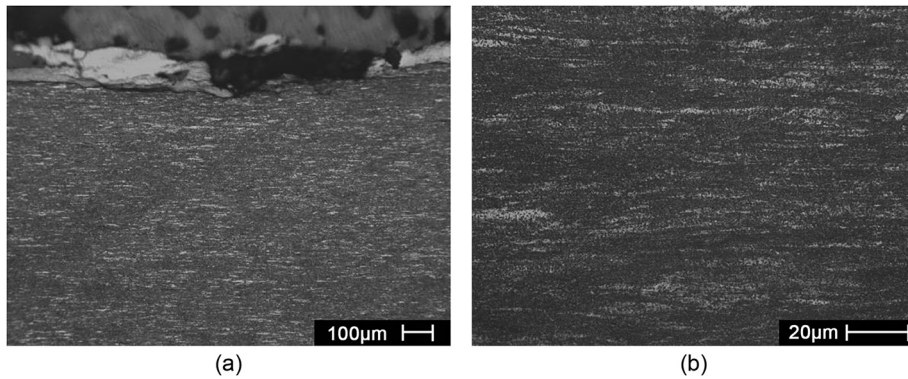


Fig. 12. Metallurgical structure of longitudinal sample of steel strand from Qishan exhumation: (a) 100 magnification, (b) 1000 magnification.

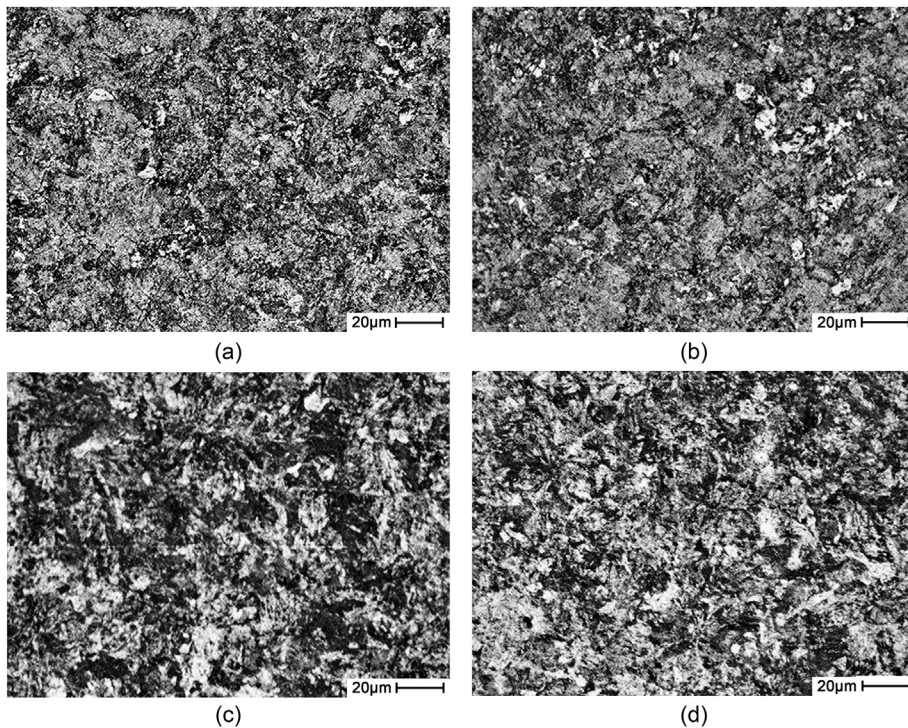


Fig. 13. Metallurgical structures of cross-sections of steel strand samples from Manwan: (a) sample 1 at a burial depth of 6 m, (b) sample 2 at a burial depth of 6 m, (c) sample 1 at a burial depth of 15 m, (d) sample 2 at a burial depth of 15 m.

and ρ is the material density. The equation shows that the relationship of corrosion rate and current density is linear. Thus it can be concluded that the sample in the pH = 12.0 solution has the lowest corrosion rate.

Curve fitting gave self-corrosion potential and corrosion current density of steel strand in different solutions (Table 16). The electrochemical test of the Shaping-II sample was conducted with a saturated calomel electrode. The potential was then converted into a standard hydrogen electrode and put into a potential-pH diagram was drawn (Fig. 18) [32]. It can be found that corrosion of the steel strand in solutions with pH of 10.9 and 12 tends to be passive, but it is close to the corrosion region when the strand is immersed into a pH 7.0 solution. More precisely, samples in a solution with 0.1% Cl^- concentration fall within the corrosion zone, which means the samples are in unstable status and liable to corrosion. The figure demonstrates that when the grout is in good quality, the anchor will be in an alkaline environment and will not corrode. Otherwise, when grouting is damaged or with poor quality, the anchor is in an environment consistent with the surrounding ground environment, which is usually neutral, and will be liable to corrosion.

4.4. Mechanical test

To understand the mechanical changes since the steel strands of Shaping-II were put into service, a microprocessor-controlled

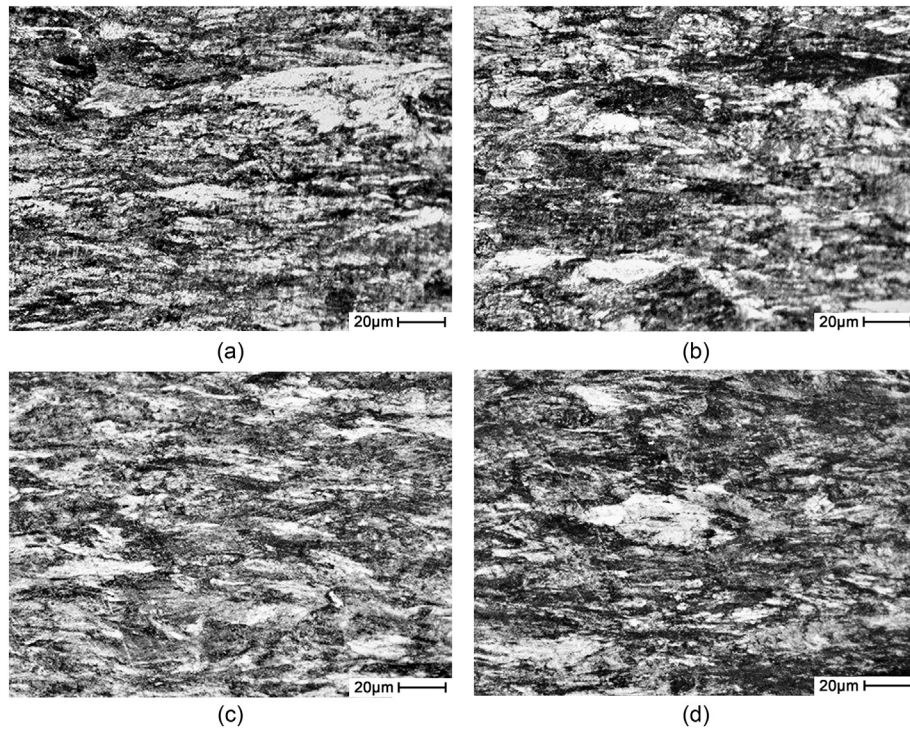


Fig. 14. Metallurgical structures of longitudinal sections of steel strand samples from Manwan: (a) sample 1 at a burial depth of 6 m, (b) sample 2 at a burial depth of 6 m, (c) sample 1 at a burial depth of 15 m, (d) sample 2 at a burial depth of 15 m.

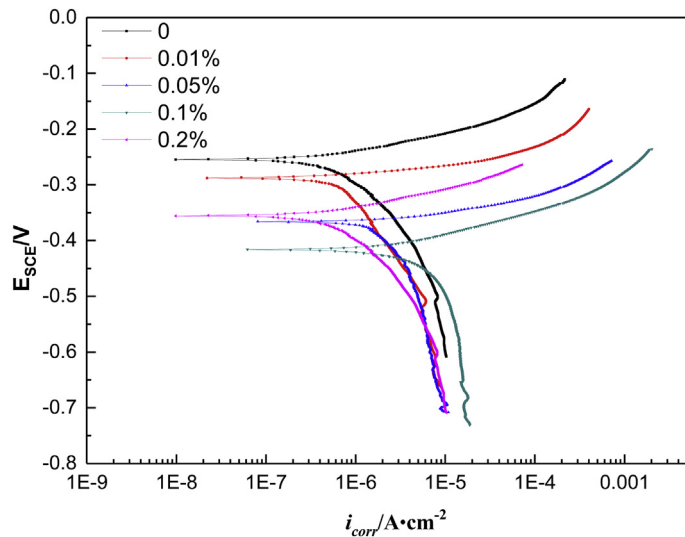


Fig. 15. Polarization curves of the steel strand from Shaping-II exhumation in neutral solution with different Cl^- concentrations.

electro-hydraulic servo all-purpose test system was used to measure tensile strength of 14 selected strands. The test results are shown in Table 17.

Among the tested steel strands, 3 out of 13 samples failed to meet requirements of the yield load and 2 of 13 samples failed to meet the requirement of elasticity modulus. The lowest yield load of the sample is 199 kN, about 14.96% lower than the standard requirement. Results of chemical composition analysis presented earlier showed that steel strands all complied with code requirements. The mechanical properties of steel strand samples were unaffected during the exhumation. As shown in Table 17, the breaking load, tensile strength and elongation of sample 1 reduced to a value lower than the standard value. This sample broke quickly after reaching the yield load, which lead to the lower breaking load, tensile strength and elongation. Considering the amount and breaking position of the wires of the samples, 10 of 14 samples broke at the clamping device and most of the samples have 1 wire broken

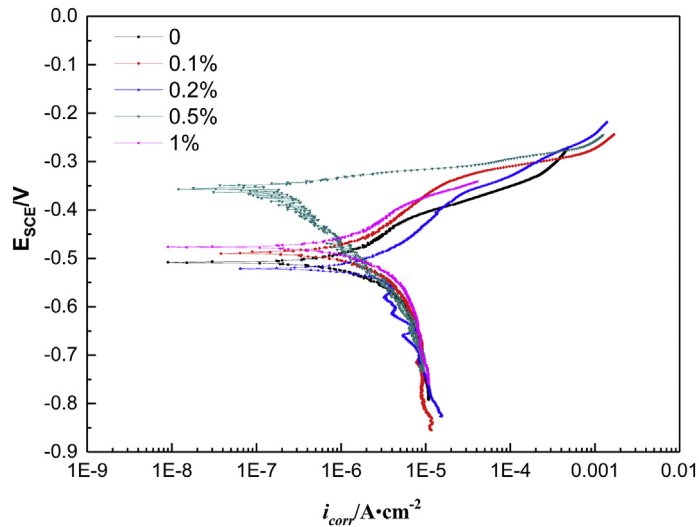


Fig. 16. Polarization curves of the steel strand from Shaping-II exhumation in pH=10.9 Ca(OH)_2 solution with different Cl^- concentrations.

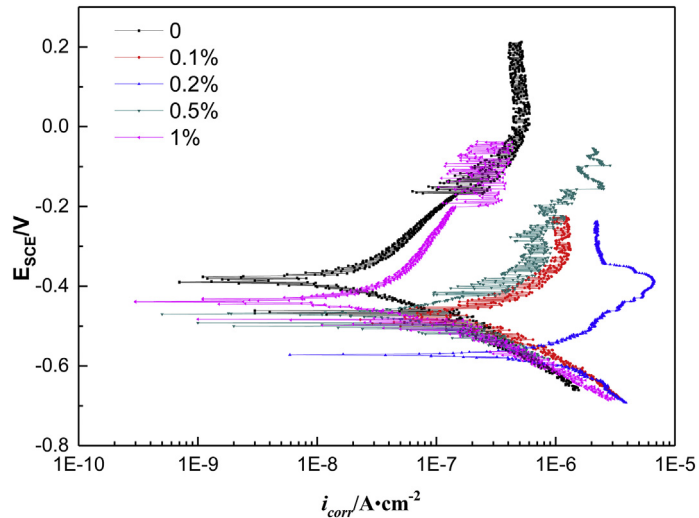


Fig. 17. Polarization curves of the steel strand from Shaping-II exhumation in saturated Ca(OH)_2 solution with different Cl^- concentrations.

Table 15

Solutions used in polarization curve test of the steel strand sample from Shaping-II.

pH	Solution	Cl^- concentration (by weight)
7.0	Pure water	0, 0.01%, 0.05%, 0.1%, 0.2%
10.9	Ca(OH)_2 solution	0, 0.1%, 0.2%, 0.5%, 1%
12.0	Saturated Ca(OH)_2 solution	0, 0.1%, 0.2%, 0.5%, 1%

during the tests, which may cause the lower value since the clamping may lead to stress concentration and the values recorded in the tests that take 1 wire broken as the whole strand fracture may be underestimated. To get the real mechanical performance of the steel strand, a better adopted clamping device would be better to ensure an even stress distribution in the wires. The tested values of the samples that broke in the middle are probable closer to the real mechanical property of the strands in service.

Despite the slight reduction in elasticity modulus, the results of mechanical tests indicate that there are no significant changes for the mechanical properties of the prestressed anchors after the anchors were put into service.

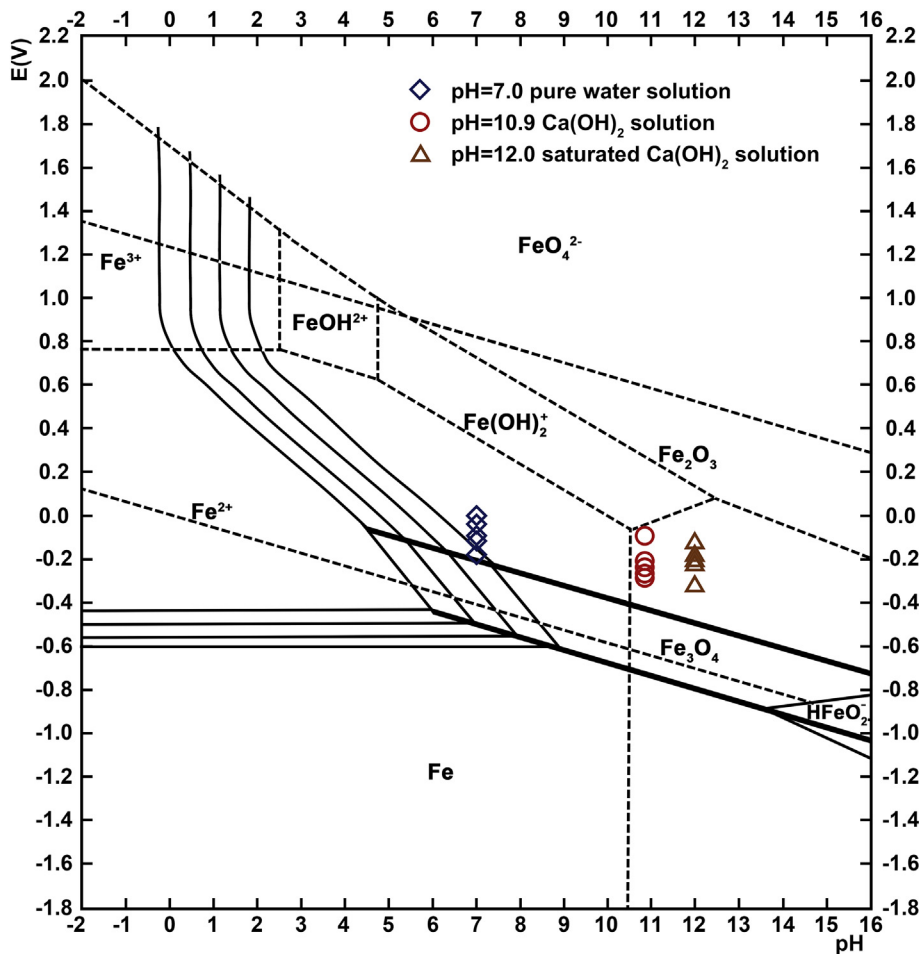
As for the steel strand samples from the Manwan site, mechanical test was conducted on 30 samples. Results are shown in Table 18.

Data in square brackets are reference values stated in the Chinese code of GB5224–85 [27]. J1, J6 and J8 are three strand samples from one anchor. This anchor consists of 8 steel strands and each steel strands are composed of 7 wires. J8 is the central steel strand

Table 16

Self-corrosion potential and corrosion current density of the steel strand sample from Shaping-II exhumation site in different solution.

pH and solution	Cl ⁻ concentration	Self-corrosion potential (V)	Corrosion current density (mA/cm ²)	E _(NHE)
pH = 7.0 Pure Water	0	-0.254	4.767×10^{-4}	-0.0125
	0.01%	-0.287	9.297×10^{-4}	-0.0455
	0.05%	-0.366	1.348×10^{-3}	-0.1245
	0.10%	-0.416	3.460×10^{-3}	-0.1745
	0.20%	-0.352	9.709×10^{-4}	-0.1105
pH = 10.9 Ca(OH) ₂ Solution	0	-0.508	1.418×10^{-3}	-0.2665
	0.10%	-0.488	5.245×10^{-3}	-0.2465
	0.20%	-0.521	8.423×10^{-3}	-0.2795
	0.50%	-0.355	6.366×10^{-3}	-0.1135
	1%	-0.476	1.128×10^{-3}	-0.2345
pH = 12.0 Saturated Ca(OH) ₂ Solution	0	-0.386	3.665×10^{-5}	-0.1445
	0.10%	-0.482	3.498×10^{-4}	-0.2405
	0.20%	-0.571	4.10×10^{-3}	-0.3295
	0.50%	-0.469	1.753×10^{-4}	-0.2275
	1%	-0.437	5.459×10^{-5}	-0.1955

**Fig. 18.** Potential-pH equilibrium diagram of Shaping-II samples.

and J1 to J6 are outside steel strands. The second digit in the sample name refers to the sample serial number started from the anchor head, and the third digit refers to the depth of the sample. It can be seen from Table 18 that after long-term loading, yield strength and tensile strength are both within the code requirements. All test values of tensile strength reported in the table are above the limit value in the code and the elasticity modulus also shows the same characteristic. With the burial depth, tensile strength and elasticity modulus do not tend to change. These characteristics seem not in consistent with the rusting distribution discovered in the exhumation test. A major reason may be the different corrosion environment for steel strands J1, J6 and J8. Steel strand J8's tensile

Table 17

Result of mechanical test on samples of prestressed steel strand from Shaping-II.

Sample	Yield load (kN) [≥234]	Breaking load (kN) [≥260]	Tensile strength (N/ mm ²) [≥1860]	Elongation (%) [≥3.5]	Elasticity modulus (GPa) [195 ± 10]	Amount and position of breaking wires
1	241.3	243.1	1737	0.3	191	1, clamping
2	239.5	269.6	1926	6.5	191	4, clamping
3	243.1	269.1	1922	6.0	189	7, mid
4	243.6	267.5	1910	5.5	198	1, clamping
5	241.8	265.8	1898	5.5	194	1, clamping
6	239.9	268.0	1914	6.0	185	1, clamping
7	235.2	263.5	1882	4.0	173	1, mid
8	–	270.7	1934	–	–	1, clamping
9	226.2	270.4	1931	7.0	185	1, clamping
10	238.2	262.4	1874	3.5	185	1, mid
11	235.7	269.9	1928	6.0	188	4, mid
12	233.5	269.4	1924	6.5	173	1, clamping
13	244.4	271.2	1937	5.5	197	1, clamping
14	199.0	265.2	1894	6.5	200	1, clamping

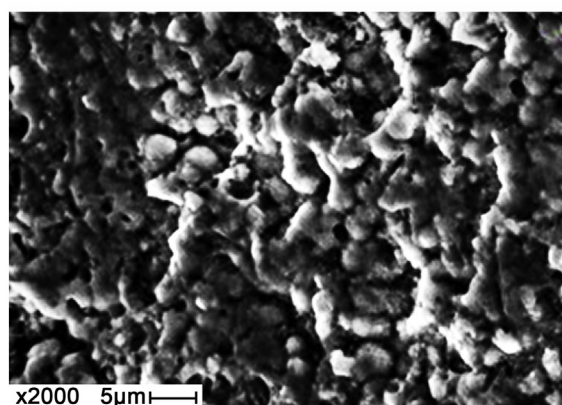
Table 18

Mechanical test results of steel strand samples from Manwan exhumation.

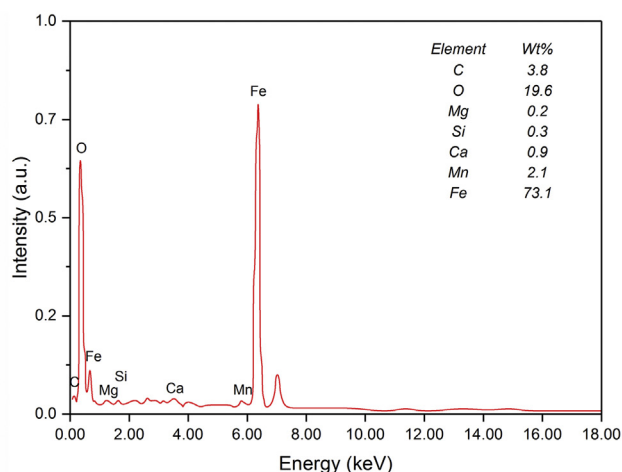
No.	Sample	Yield load (kN) [≥174.83]	Breaking load (kN) [≥219.52]	Tensile strength (MPa) [≥1470]	Elongation (%) [≥3.5]	Elasticity modulus (GPa)
1	J1-01-1.20		213.06	1520	5.9	194
2	J1-02-2.35		215.12	1540	7.3	183
3	J1-03-3.50		215.51	1540	7.2	188
4	J1-04-4.65		214.22	1530	6.7	191
5	J1-B1-8.10		215.35	1540	7.0	192
6	J1-B2-9.25		213.60	1530	6.2	187
7	J1-B3-10.4		215.91	1540	7.6	189
8	J1-07-19.6		216.55	1550	7.2	195
9	J1-08-20.75		215.75	1540	7.4	193
10	J1-09-21.85		215.97	1540	7.5	191
11	J6-01-1.20	178.50	219.40	1570	7.0	195
12	J6-02-2.20	180.50	215.80	1540	6.0	190
13	J6-03-3.20	174.70	214.90	1535	6.5	185
14	J6-04-4.20	180.20	216.30	1545	6.5	188
15	J6-08-8.20	176.80	218.50	1560	7.0	194
16	J6-09-9.20	178.50	218.50	1560	6.5	199
17	J6-10-10.20	176.10	221.40	1580	7.0	192
18	J6-19-19.20	186.30	221.90	1590	8.5	200
19	J6-20-20.20	186.00	221.40	1580	8.0	203
20	J6-21-21.20	183.60	219.40	1570	6.5	204
21	J8-01-1.20	181.20	220.90	1580	7.0	204
22	J8-02-2.20	177.80	220.90	1580	7.5	204
23	J8-03-3.20	178.50	219.40	1570	6.5	191
24	J8-04-4.20	178.00	220.90	1580	7.5	201
25	J8-07-7.20	179.50	221.40	1580	7.0	205
26	J8-08-8.20	177.50	220.90	1580	7.5	199
27	J8-09-9.20	180.40	221.40	1580	7.0	200
28	J8-18-18.20	187.50	221.90	1590	7.0	195
29	J8-19-19.20	187.90	220.90	1580	7.0	202
30	J8-20-20.20	186.30	220.90	1580	7.0	196

strength has minor changes along with burial depth, with an average value around 1580 MPa. J1 and J6 follow approximately the tendency that tensile strength increases with increasing burial depth. There is a gradual increase tendency from orifice to free stressing length and from direction turning point of borehole to bonded length. The tensile strength declines slightly at the direction turning point of the borehole. In addition, the tensile strength of J1 is less than that of J6 and J8 overall and the tensile strength of J6 is lower than that of J8. This distribution coincides with the analytical result of the anchor exhumation.

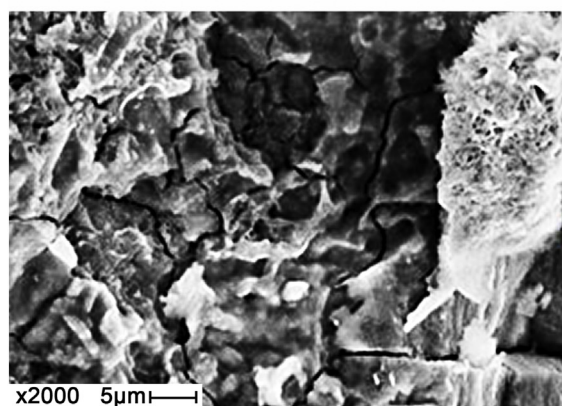
Steel strands selected for the mechanical test from the Manwan exhumation site are specific. J8 is in the center of the anchor and is only slightly corroded thanks to the protection from mortar inclusion and surrounding steel strands. This explains why its tensile strength remains basically unchanged from the orifice to the bonded length. J6 is the external steel strand that affected by the surrounding environment easier. The breaking load and tensile strength of the bonded length are generally higher than that of the unbonded length of the strand. This may be caused by the corrosion thanks to the environmental effects. J6 was found in a slack steel strand at the beginning of the anchor exhumation. This may be due to the tensioning method during the construction that the steel



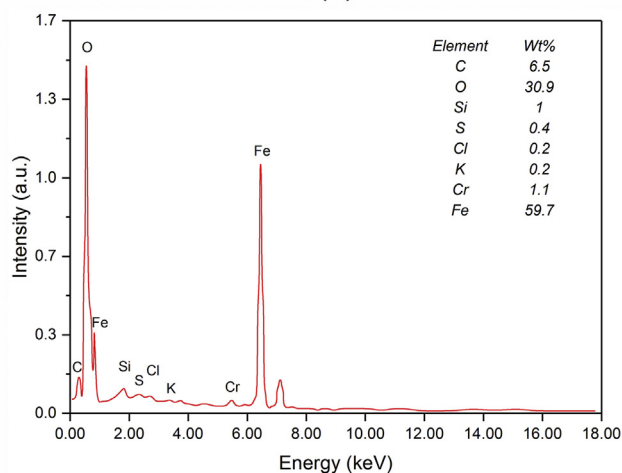
(a)



(b)



(c)



(d)

Fig. 19. Morphology and EDS analysis of corrosion byproducts at the crack initiation point of Qishan sample: (a) microscopic morphology of test point 1, (b) EDS result of test point 1, (c) morphology of test point 2, (d) EDS result of test point 2.

strands were tensioned one by one in order and when it came to the last steel strand, previous strands had been loose. J1 is in line with the outside with a high prestressed which was found corrode seriously. Compared to J6, the high stress may contribute to the corrosion besides the surrounding environment. This may be the reason why J1 has the lowest tensile strength.

In summary, mechanical properties of the exhumated anchor still meet the code requirements after 20 years' service. Both the high stress in the steel strands and the surrounding corrosion environment are important factors of the corrosion of steel strands.

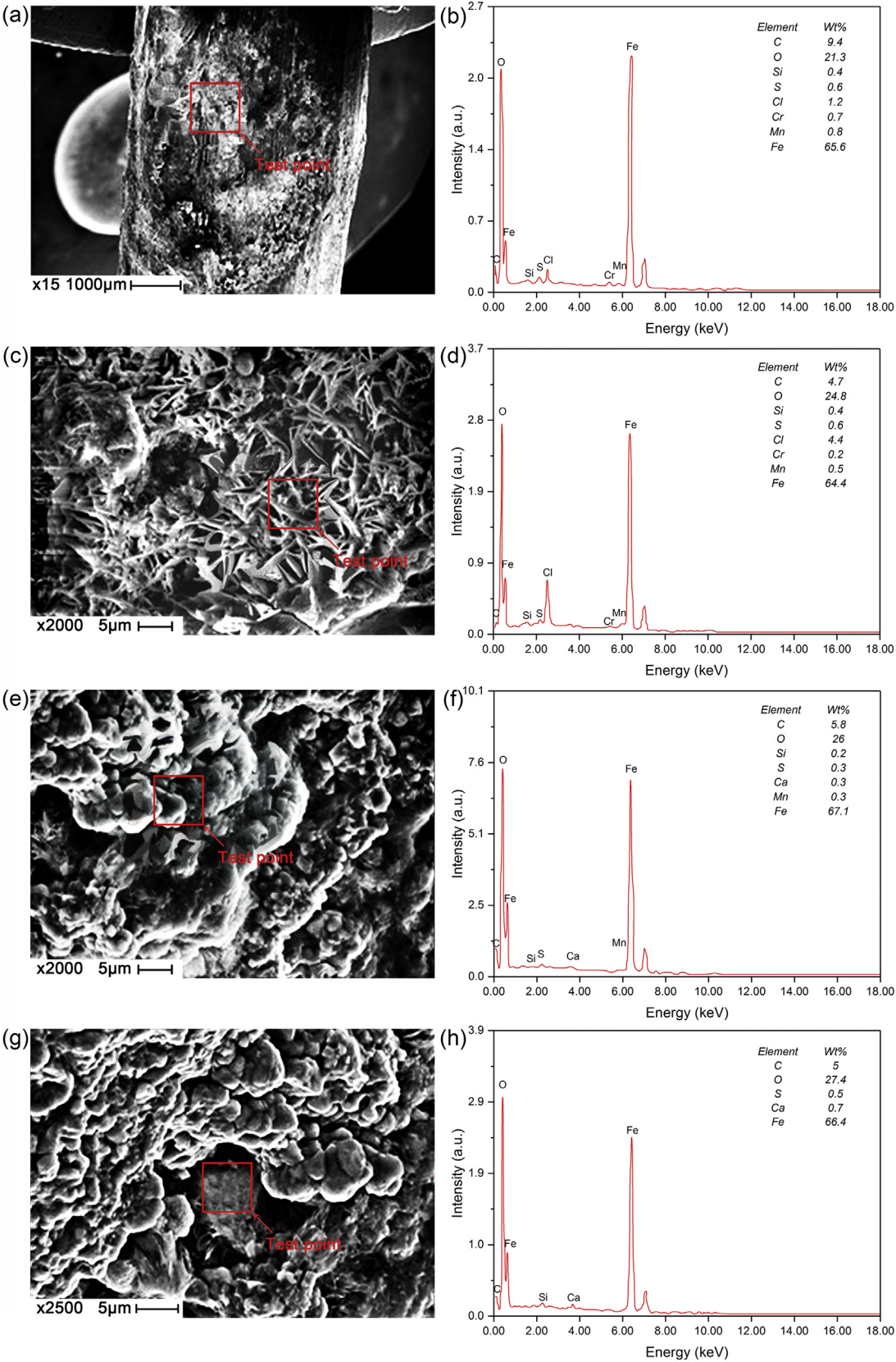
4.5. Analysis of corrosion byproducts

4.5.1. Corrosion byproducts of Qishan samples

Macroscopic and microscopic morphology were studied after the fracture sample (Fig. 6) was cleaned. Cleaning was conducted until the fracture surface was judged by the naked eye to be covered by blackish-yellow corrosion byproducts. The fracture is an oblique fracture with a longitudinal length of 25 mm. Microscopic morphology and EDS analysis of corrosion byproducts at the crack are shown in Fig. 19. According to the analysis, the surface is largely covered by iron oxides with sulfur-containing corrosion byproducts in localized areas.

Microscopic morphology and EDS analysis of corrosion byproducts on the surface near to the fracture are shown in Fig. 20. Corrosion byproducts at this location seriously peeled off. The results show that the byproducts are composed of Fe, O, Cl, S, Si, Mn and Cr. In local areas, the percentages of Cl and S reached 4.4% and 0.6%. The fracture surface of the anchor is heavily corroded with some corrosion products peeled off in blocks. Corrosion may be largely caused by O, Cl and S.

A metallurgical sample with transversal section taken from a concave corrosion area 90 cm away from the anchor head was polished. Morphologic and EDS analysis was conducted. The results are presented in Figs. 21 and 22. Fig. 21 displays a low multiplier



(caption on next page)

Fig. 20. Morphology & EDS analysis of corrosion products on the surface near the fracture of Qishan sample: (a) microscopic morphology of test point 1, (b) EDS result of test point 1, (c) microscopic morphology of test point 2, (d) ED results of test point 2, (e) microscopic morphology of test point 3, (f) EDS result of test point 3, (g) microscopic morphology of test point 4, (h) EDS result of test point 4.

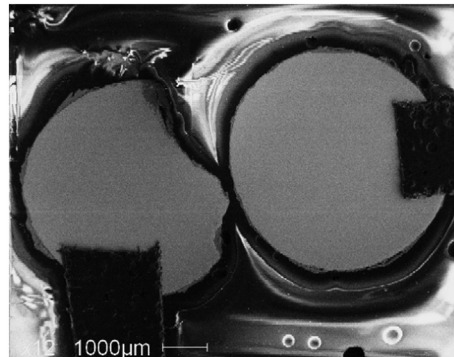


Fig. 21. Low multiplier photograph of metallurgical transversal section of Qishan steel strand samples with and without pit.

photo of a steel wire fracture. Fig. 22 show microscopic morphology and corrosion byproducts composition analysis at different locations of a transversal section sample with a pit. Fig. 23 displays microscopic morphology and corrosion byproducts composition analysis of a steel strand transversal section sample without a pit. Results indicate several pits are 0.9 mm in depth. In areas with pitting corrosion, areas near the body and the surface of the steel strand are all covered with corrosion byproducts that contain O, Cl, S or O and S. In areas with no pitting corrosion, areas near the surface and the body of the steel strand are covered with O containing corrosion byproducts.

4.5.2. Corrosion byproducts of Manwan

Only two typical morphology, yellow rust and a bluish-grey substance, are observed on the surface of the steel strands of Manwan samples. The former can be judged easily as common corrosion byproducts, but the latter is hard to be determined because of the relatively smooth surface. Both substances were manually sampled at different locations and performed X-ray Diffraction (XRD) testing.

Main components of the bluish-grey and yellow rust were inferred through comparison between standard calorie and diffracted intensity of all peaks in the XRD diffractogram (Figs. 24 and 25).

The bluish-grey substance is found to consist of FeO, FeOOH and some $\text{Fe}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$ while yellow rust is composed of Fe_3O_4 , Fe_2O_3 and some FeOOH. Both substances are therefore iron oxides. The bluish-grey color is mainly from the FeO considering that the FeO is black and FeOOH is yellow. The FeO is unstable and easy to be oxidized to form Fe_3O_4 . Fe_3O_4 is black, so the yellow rust should be Fe_2O_3 and FeOOH primary. It can be conjectured that the strands in the environment corroded at the first and after that the corrosion continued to carry on because of the oxidation caused by the oxygen in the surrounding environment. The bluish-grey substance and yellow rust are products of two different corrosion stage that developed with the sustained environmental effects.

4.6. Microhardness analysis

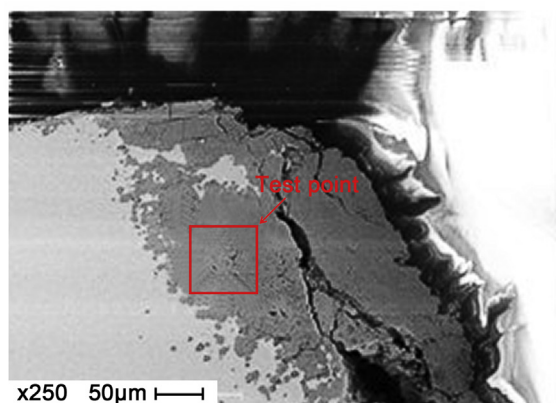
According to the microhardness analysis on the longitudinal metallurgical sample of Qishan steel strand, the average value is 469.3 HV0.2. Tensile strength as estimated from hardness test results and the Conversion of Hardness and Strength for Ferrous Metal (GB/T 1172) [33] is 1550 MPa. This is significantly higher than the stress generated by lock-off load (120 kN). This further supports the direct correlation between premature fracture and corrosion of this anchor.

4.7. Grout testing

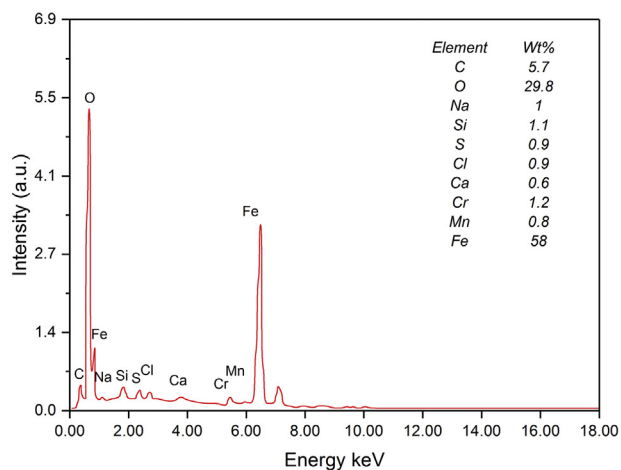
Manwan anchor samples display grout solidification failure in an ongoing 20 years process. Compared to a traditional cement mortar environment, the free stressing length of the tested anchor at the Manwan site is composed of neat cement grout and located in a well-defined closed cavity. Studies were carried out on the material composition, microstructure and closed environment to explain the causes of this failure.

The exhumed grout is an ordinary silicate cement that consists of tricalcium silicate, dicalcium silicate and tricalcium aluminate ($3\text{CaO} \cdot \text{SiO}_2$, $2\text{CaO} \cdot \text{SiO}_2$, $3\text{CaO} \cdot \text{Al}_2\text{O}_3$). XRD analysis was conducted on the unsolidified grout samples of the free stressing length and the cement mortar sample of the bond length from the Manwan anchor exhumation (Figs. 26 and 27).

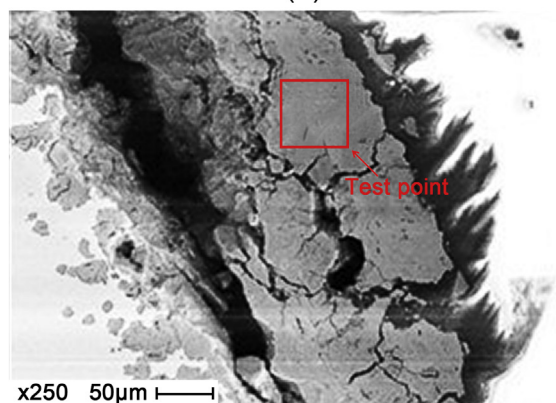
Comparison between diffracted intensity peaks of the XRD diffractograms and the standard reference pattern [34] proved that the grout of free stressing length and bond lengths are basically of same composition, including tobermorite gel, portlandite crystals (calcium hydroxide with formula $\text{Ca}(\text{OH})_2$), ettringite crystals (hydrous calcium aluminum sulfate with formula



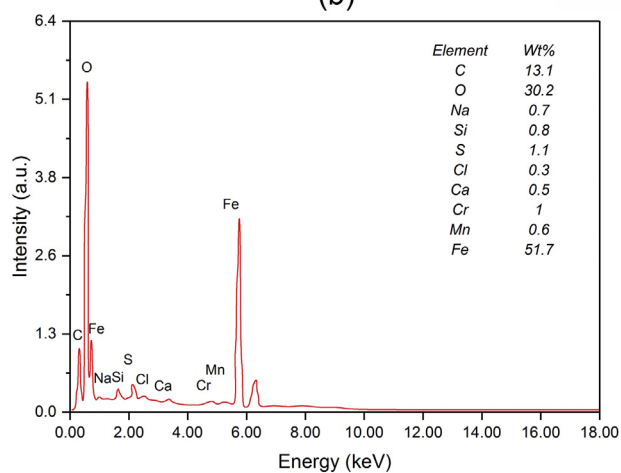
(a)



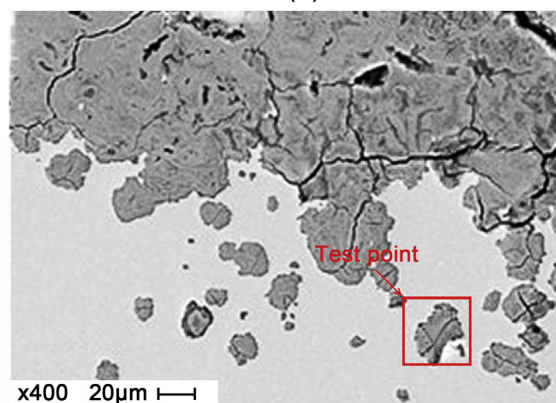
(b)



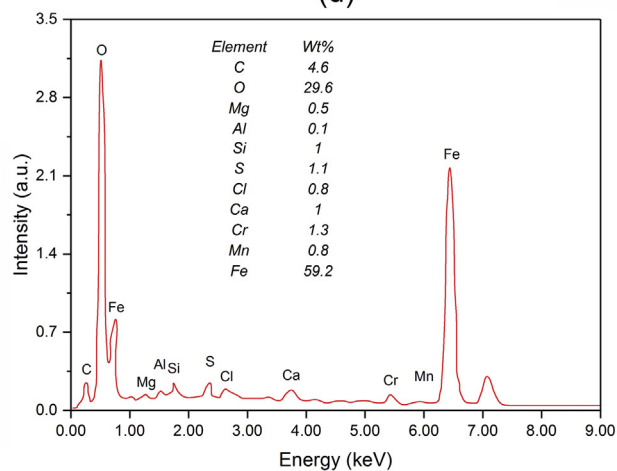
(c)



(d)



(e)



(f)

Fig. 22. Morphology and EDS analysis of corrosion byproducts of metallurgical transversal section sample with pit of steel strand from Qishan: (a) microscopic morphology near the steel strand body, (b) EDS result near the steel strand body, (c) microscopic morphology on the surface of the corrosion byproducts, (d) EDS result on the surface of the corrosion byproducts, (e) microscopic morphology inside the pit, (f) EDS result inside the pit.

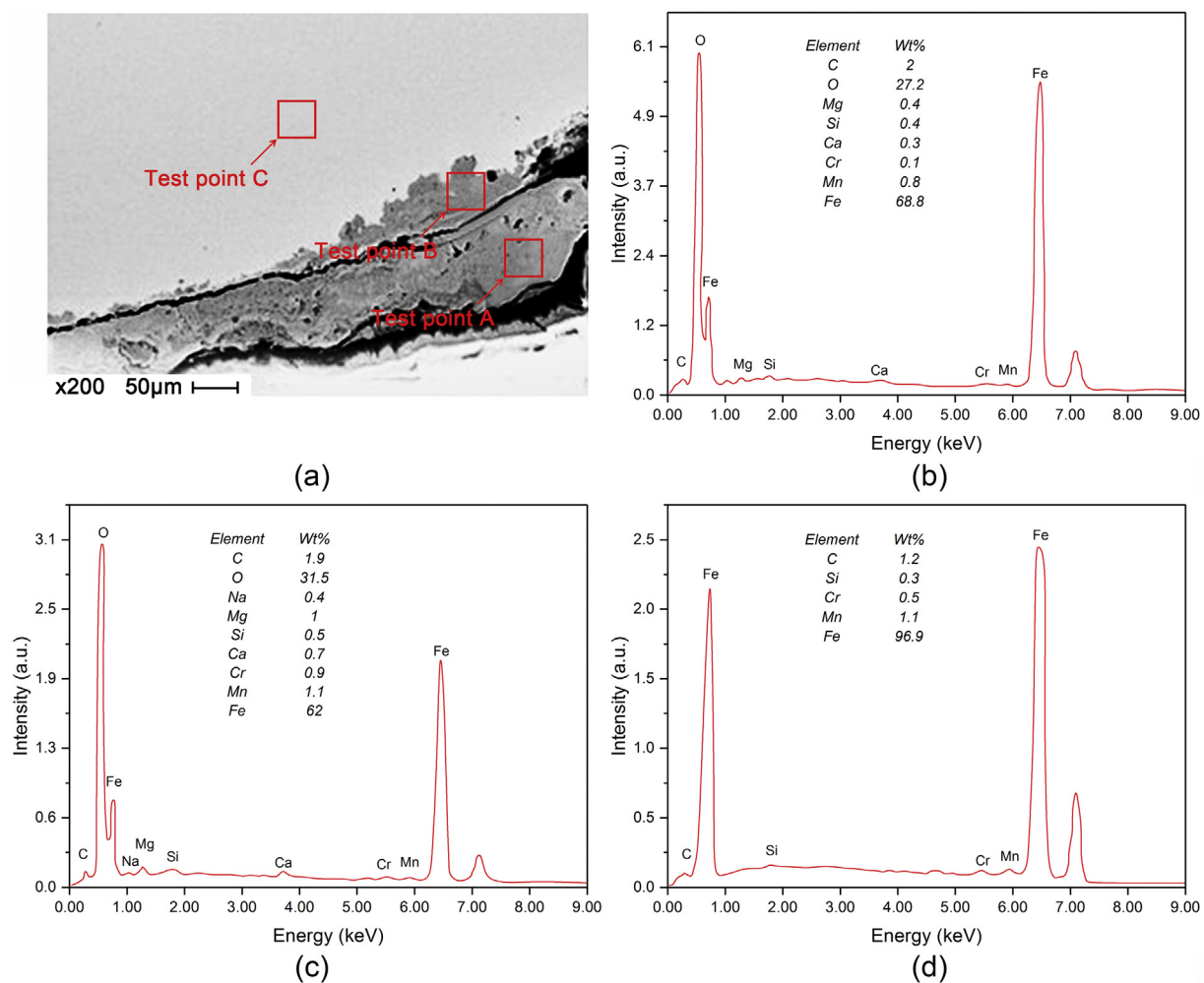


Fig. 23. Morphology and EDS analysis of corrosion byproducts of metallurgical transversal section sample without pit of steel strand from Qishan: (a) morphology of the sample, (b) EDS result of corrosion byproducts at the surface (test point A), (c) EDS result of corrosion byproducts near the steel strand body (test point B), (d) EDS result of the steel strand body (test point C).

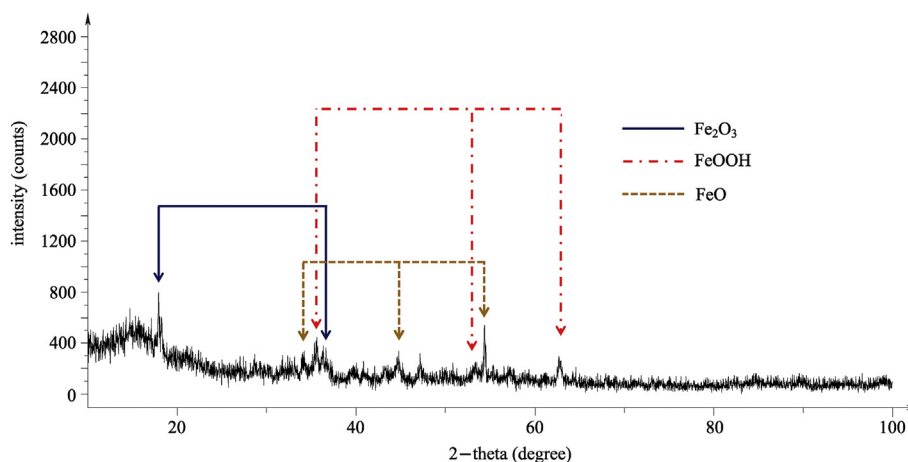


Fig. 24. XRD diffractogram of bluish-grey rust on Manwan sample. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

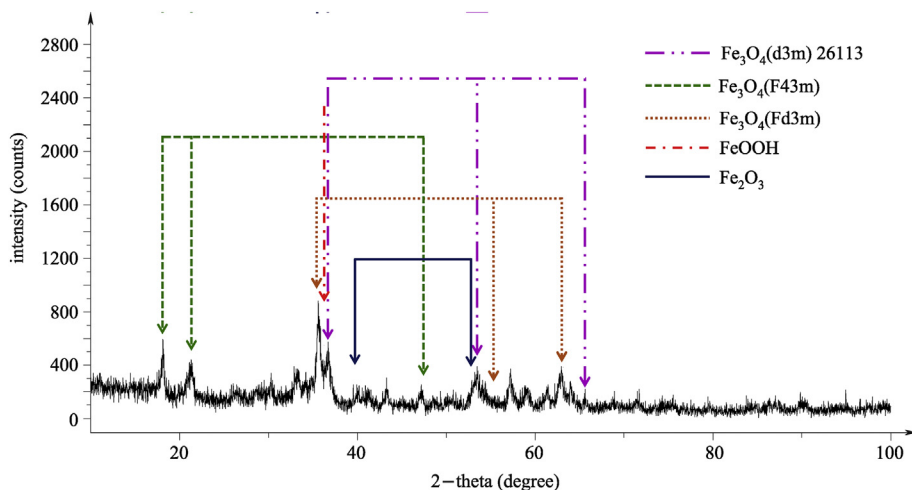


Fig. 25. XRD diffractogram of yellow rust on Manwan sample.

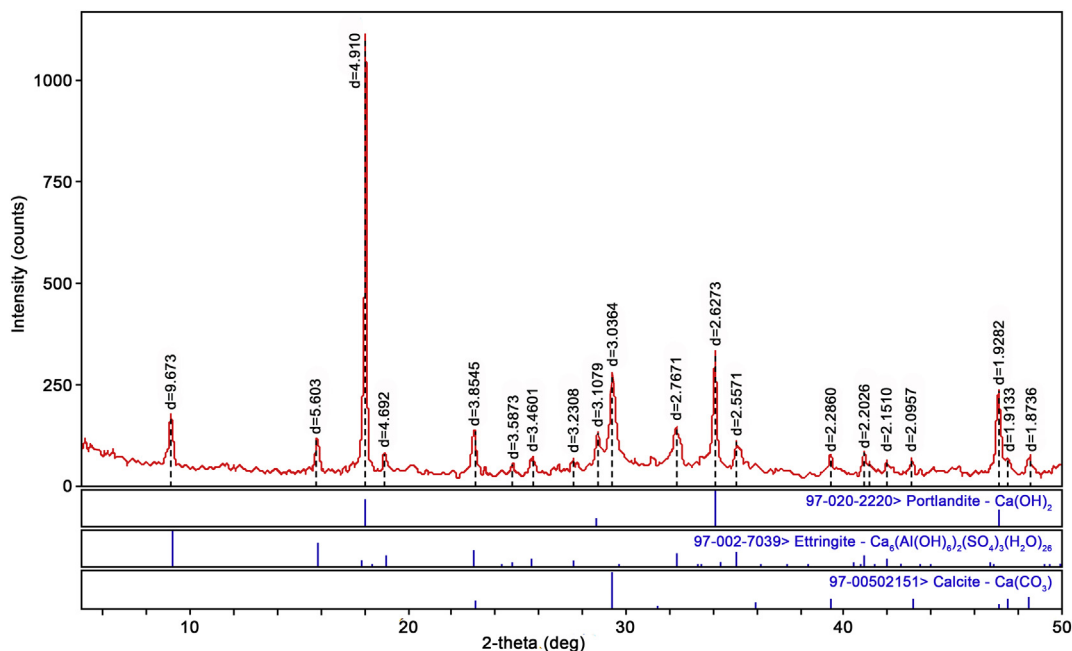


Fig. 26. XRD analysis of grouting on free stressing length of Manwan sample.

$\text{Ca}_6\text{Al}_2(\text{SO}_4)_3(\text{OH})_{12}\cdot 26\text{H}_2\text{O}$) and a small amount of calcite ($\text{Ca}(\text{CO}_3)$) crystals.

Grout microstructures of free stressing length and bond length observed through a scanning electron microscope are presented respectively in Fig. 28 and Fig. 29. Comparing Fig. 28(b) and Fig. 29(a), under 1500 magnification, the grout microstructure of the free stressing length is more porous than that of the bond length.

As shown in Figs. 28 and 29, microstructures of grout in the free stressing length and in bond length show two obvious characteristics. First, hydrated hardened products of unsolidified grouting of the free stressing length is mainly formed of ettringite and of other acicular crystals with a small amount of tabular portlandite. Hydrated minerals of the grout in bond length include tabular portlandite and a small amount of acicular ettringite. Second, crystals and colloids of the free stressing length are not closely cemented. Instead, they are loosely packed and in the shape of flakes. Minerals of the grouting in bond length are more tabular crystals of portlandite with interspersed ettringite and other acicular crystals filled by colloidal hydrated calcium silicate. These are well cemented and compacted.

5. Assessment of anchorage structures

The concealment and the complexity of the environment make the assessment of the safety and longevity of anchorage structures

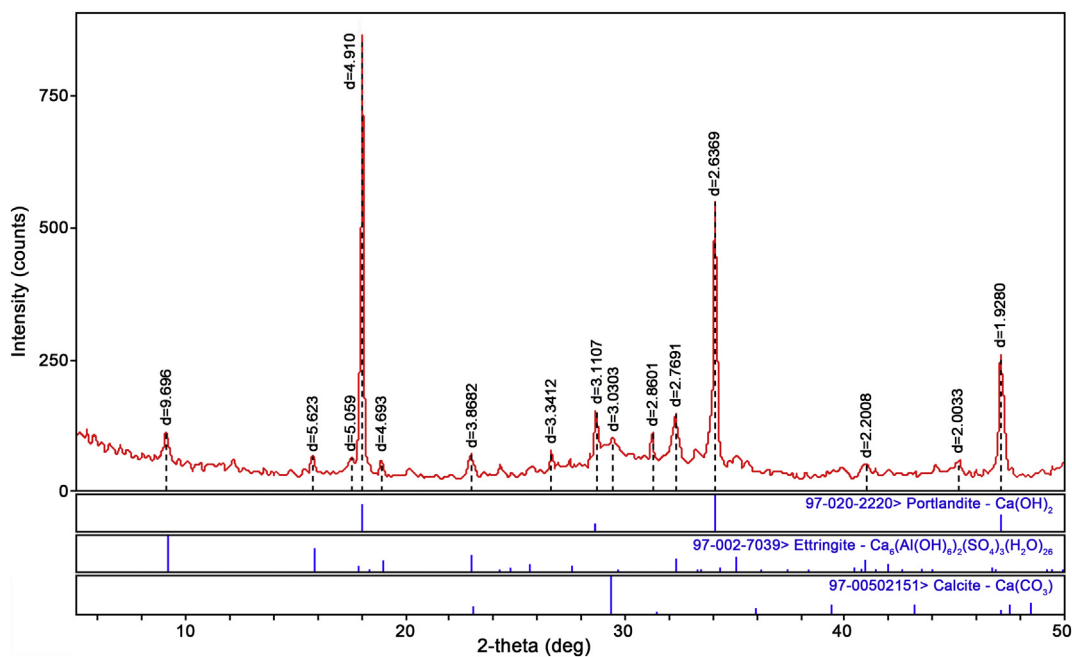


Fig. 27. XRD analysis of grouting on bond length of Manwan sample.

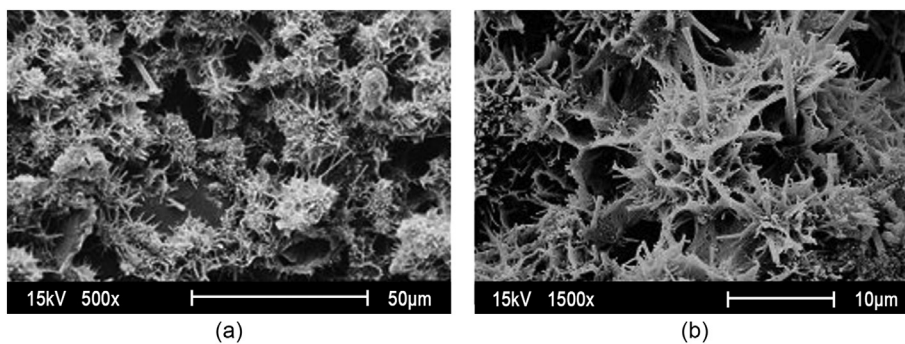


Fig. 28. SEM photographs of grout on free stressing length of Manwan sample: (a) 500 magnification, (b) 1,500 magnification.

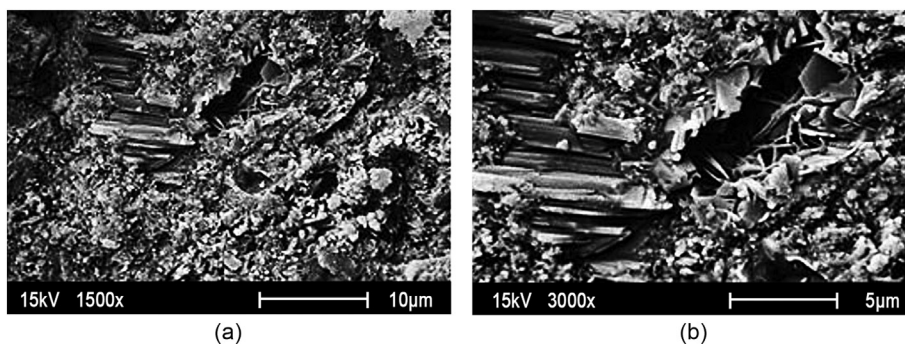


Fig. 29. SEM photographs of the grout on bond length of Manwan sample: (a) 1500 magnification, (b) 3000 magnification.

that still in service a difficult work. The works performed to evaluate the working state and durability in the field must be non-destructive. This may make the work even harder. Study of failure cases is a direct and effective method that can truly reflect the failure types and mechanism. However, compared to the extensive application of the anchor structure, the failure cases are few and rarely reported. Furthermore, environment and geology of anchorage projects are different sharply. How to use the results of failure case analysis in a broad way is also an unanswered question.

This study performed an environment investigation, including damaging ions in groundwater, pH of the groundwater, temperature and humidity. The effects of these factors were evaluated and discussed. The results may be used in the future work of effects assessment of environmental factors on the corrosion of anchors and helpful in identification of critical factors.

The tests in the field include underground stray current test and morphologic characteristics analysis. The stray current test results discovered that the stray current may not be the key factors in those projects. Unlike the traffic tunneling, for similar projects, the test of stray current in the future works may not be necessary. The morphologic characteristics analysis provided the corrosion dimension and distribution. The results are useful in determination of the locations that are easy to corrode due to the environment attack and the structure flaws.

More laboratory tests were conducted to help to capture the effects of the corrosion on the anchors. The failure mechanism was also studied based on these tests. The results are helpful to understand the failure type of the anchor structure and make clear the key points that should be paid more attentions to in the assessment works of anchorage systems.

The test methods applied in the study and the tests results and analysis may be considered as a system that can be used as reference or comparison in related works.

6. Discussion

Investigations of environmental conditions and corrosion morphology of steel bars and strands were performed in four field exhumations. Samples were taken from these sites and analyzed in laboratory.

Groundwater samples from the four exhumation sites were tested. Based on the code, test results indicate that the pH and deleterious ions of groundwater impose little influence on the corrosion of anchors in-service. However, the reference values suggested by the codes can only help to evaluate general corrosion of anchors due to pH and deleterious ions of groundwater, and their long-term influence is not considered. In addition, no consideration is given to the combined effect of groundwater, deleterious ions and other corrosion factors in practice.

Stray current test was conducted at Jinping-I site. The potential difference between anchors and rock mass slightly fluctuates. Testing results indicate there is no obvious stray current around the anchors.

According to typical corrosion theories, both temperature and humidity impose important influence on iron and steel corrosion [24]. Temperature and humidity were tested at Qishan and Jinping-I sites. At Jinping-I, temperature and humidity changes were small. While at Qishan, essential changes were observed in temperature at five locations. Nevertheless, it is difficult to evaluate from field tests, the effect of temperature and humidity on corrosion of anchors. A major reason is perhaps that the geological environment of an anchor is relatively unique and the effect of temperature and humidity cannot be separated independently from other factors.

Undoubtedly, visual examination of morphological characteristics is an easy and the most direct method to assess the degree, location and type of anchors corrosion. In addition, it allows a preliminary evaluation on corrosion byproduct composition through color, distribution and other directly observable characteristics.

In laboratory, chemical composition of strand samples from Shaping-II and Manwan sites was determined through chemical component analysis. The measured chemical composition for the anchors working for 3 (Shaping-II) and 20 years (Manwan) showed no obvious change and still satisfied the current code requirements. Tests results also indicate that the superficial corrosion and high stress have no essential effect on the chemical composition of steel strands.

Observations of metallurgical structures of samples from Shaping-II, Qishan and Manwan denote no change in the metallurgical structures of anchors in service.

Corrosion resistance and the corrosion rate of a steel strand from Shaping-II site were tested through analysis of polarization curves in solution with different pH values and Cl^- concentrations. For anchors existed in an alkaline environment, corrosion tendency was found passive and the steel strand did not corrode. If grout is damaged or of bad quality, anchors are exposed to surrounding neutral ground environment and is subject to corrosion. This explains why the bond length wrapped with good quality of grout corroded much moderately. Other sections of anchors where corrosion protection measures and grout have been partly or fully damaged were easily corroded.

Mechanical tests at Shaping-II and Manwan sites indicate no change in mechanical performance of prestressed anchors in service. Considering the short service time of Shaping-II samples and the lower corrosion of Manwan samples, it is reasonable to state that the mechanical performance remains qualified.

Analysis of corrosion byproducts may help to determine the type of corrosion for in service anchors and dominant corroding factors. EDS analysis and morphology of corrosion byproducts on the fracture of the Qishan sample and its surroundings suggest that ambient O, Cl and S have serious influence on the corrosion. Pitting corrosion around the fracture contains O, Cl and S byproducts. Other locations present O containing corrosion byproducts. Conclusions may be acquired based on the tests results that the participation of deleterious ions can encourage the development of corrosion that may even lead to structural damage. In Manwan, XRD analysis on a test sample indicates corrosion byproducts are mainly iron oxides, including FeO, FeOOH, Fe_3O_4 and Fe_2O_3 , with no involvement of deleterious ions. The latter may be explained why there is only moderate corrosion of anchor studied at this site.

Microhardness tests performed on the longitudinal metallurgical structure of a steel strand sample from Qishan estimate that tensile strength would be of 1550 MPa which is far more than the actual working stress. This may lead to the possibility that the fracture of the steel stands may owing to the corrosion.

7. Conclusion

Four field exhumations are implemented to evaluate the state of corrosion and working performance of anchors in service through field investigations and laboratory tests. The investigations and tests results suggest the following conclusions:

1. Analysis of metallurgical structure and polarization curve shows that chemical performance of prestressed anchors displays no essential change and thus remains relatively stable in complex ground environments.
2. Mechanical tests on anchor samples also reveal no essential change in chemical performance of prestressed anchors in service.
3. On-site analysis of morphological characteristics illustrates a progressive decline in corrosion from anchor head, free anchor length to the bond length, which indicates that anchor head is most liable to corrosion and the bond length exhibits the best corrosion resistance properties.
4. Analysis of corrosion byproducts indicates that the presence of deleterious ions can accelerate the initiation and development of corrosion, although the content of deleterious ions in the environment does not exceed the code requirements.
5. Stray currents in the field do not impose an obvious influence on the corrosion of anchors.

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Declarations of interest

None.

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